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Review papers

Applications of deep learning in water quality management: A state-of-the-art review

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ABSTRACT

Excellent water quality (WQ) is an indispensable element in ensuring sustainable water resource development. It is highly associated with the 3rd (good health and well-being), the 6th (clean water and sanitation), and the 14th (life below water) listed items of the United Nations' Sustainable Development Goals. Thus, policymakers have always been seeking strategies to manage WQ efficiently. Recent advancements in computational technologies have created enthusiasm for using artificial intelligence, particularly deep learning (DL), in WQ management. This review provides a comprehensive overview of the application of DL in WQ management, covering developments from 2011 to 2022, in maintaining the temporal relevance of this review. In this paper, a brief description of different variants of DL models, including the recurrent neural network (RNN), long short-term memory network (LSTM), convolutional neural network (CNN), etc, are presented. The distinct approaches in the optimization, hybridization and relevant data pre-processing techniques suitable for the DL models, are also discussed. This is the first review paper that extensively discusses the application of DL models for forecasting WQ parameters in various water bodies, such as rivers, lakes, coastal areas, etc. The emergence of the Internet of Things (IoT) and cloud computing that revolutionized DL approaches in WQ management are also presented. This review paper serves as a complete easy guideline for the researchers in the field of DL-based WQ management. The findings of this review paper may help policymakers to enhance their decision-making process with the hope that regional environmental welfare can drastically be improved.

1. Introduction

The management and regulation of water quality (WQ) are worldwide challenges since the WQ is affected by both local processes (e.g., point source pollution) and global processes (weather patterns). Some of these involve human decisions that are difficult to predict or manage (e.g., the use of fertilizers) while some are beyond that of a manager's control (e.g., accidental events) (de Paul Obade and Moore, 2018; Salerno et al., 2018; Kachroud et al., 2019; Destandau and Zaiter, 2020). Vast efforts have been made to resolve the issues related to WQ management, including real-time monitoring, WQ parameters modelling and forecasting based on computer simulation or artificial intelligence (AI). These efforts are aligned well with the 6th Sustainable Development Goal (clean water and sanitation) as endorsed by the United Nations. Over the past decade, there has been growing interest amongst the scientific community to apply machine learning (ML) or AI tools for modelling WQ. This evolution is proven through the increase in the number of Scopus-indexed publications on managing WQ using ML or AI tools. Fig. 1 shows the results of this comprehensive statistical study using the Scopus search engine over the past decade. To obtain the number of publications discussing the application of ML or AI for WQ management, a paper search was done on the Scopus database using several keywords including "machine learning", "artificial intelligence" and "water quality" from 2011 to 2022. Similarly, the search term "deep learning" was included to obtain the number of publications that had applied DL models for WQ management. It can be seen that since 2016, the number of research publications concerning WQ management using AI or ML has been increasing annually. This observation is in agreement with the increase in the proportion of studies using DL models in WQ-related research works. In other words, the research field is experiencing tremendous growth with the improvement of computing facilities alongside the development of various technologies and algorithms.

The potential of ML of AI in WQ management, particularly within the aspect of decision-relevant prediction has been proven (Varadharajan et al., 2022). Numerous studies had been conducted to investigate the

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Nomenclature

Abbreviations

3D	Three-dimensional
AE	Autoencoders
AI	Artificial intelligence
AN	Ammoniacal nitrogen
ANFIS	Adaptive neuro-fuzzy inference system
ANN	Artificial neural network
ARIMA	Autoregressive integrated moving average
Bi-GRU	Bi-directional gated recurrent unit
Bi-LSTM	Bi-directional long short-term memory
Bi-RNN	Bi-directional recurrent neural network
BOD	Biochemical oxygen demand
BPNN	Back-propagation neural network
CEEMDAN	Complete ensemble empirical mode decomposition with adaptive noise
CNN	Convolutional neural network
DBN	Deep belief network
DL	Deep learning
DO	Dissolved oxygen
DT	Decision tree
EC	Electrical conductivity
ED	Encoder-decoder
EEMD	Ensemble empirical mode decomposition
EFDC	Environment Fluid Dynamics Code
EL	Event-triggered learning
ELM	Extreme learning machine
EMD	Empirical mode decomposition
ESN	Echo state network
GA	Genetic algorithm
GBDT	Gradient boosting decision tree
GBRT	Gradient boosted regression tree
h	Hour
HOPLS	Higher-order partial least squares
IMF	Intrinsic mode function
IoT	Internet of things
KNN	K-nearest neighbours
LR	Linear regression
LSSVM	Least-squares support vector machines
LSSVR	Least squares support vector regression
MAE	Mean absolute error
ML	Machine learning
MLP	Multilayer perceptron
MT	Multi-task learning
NSE	Nash-Sutcliffe efficiency
PCA	Principal component analysis
PG	Process-guided
PSO	Particle swarm optimization
R^2	Coefficient of determination
RBF	Radial basis function
RF	Random forest
RGN	Recurrent graph network
RMSE	Root mean squared error
RNN	Recurrent neural network
SAE	Sparse autoencoder
SARIMA	Seasonal autoregressive integrated moving average
SG	Savitzky-Golay
SMOTE	Synthetic minority oversampling technique
SRU	Simple recurrent unit
SSS	Sea surface salinity
STVAR	Space-time autocorrelation
SVM	Support vector machine
SVR	Support vector regression
TCN	Temporal convolutional network
TDS	Total dissolved solids
TL	Transfer learning
TN	Total nitrogen
TP	Total phosphorus
TSS	Total suspended solids
VMD	Variational mode decomposition
WQ	Water quality
WQI	Water quality index
WWTP	Wastewater treatment plant

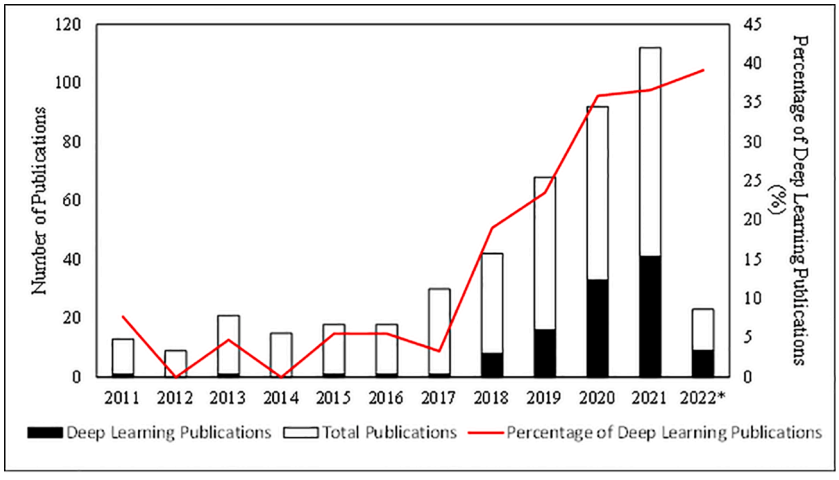


Fig. 1. Number of Scopus-Indexed Publications Related to the Application of ML or AI for WQ Management from 2011 to 2022* (* as of the date of preparation of this manuscript).

feasibility, practicability and efficiency of ML or AI in WQ management, while several review articles had been published to summarize and criticize the available literature. Despite this, the authors are still eager to build on the previous reviews in the literature as there is insufficient coverage in certain key areas. For instance, Zhao et al. (2020) only focused on reporting the utilization of AI in the operation of wastewater

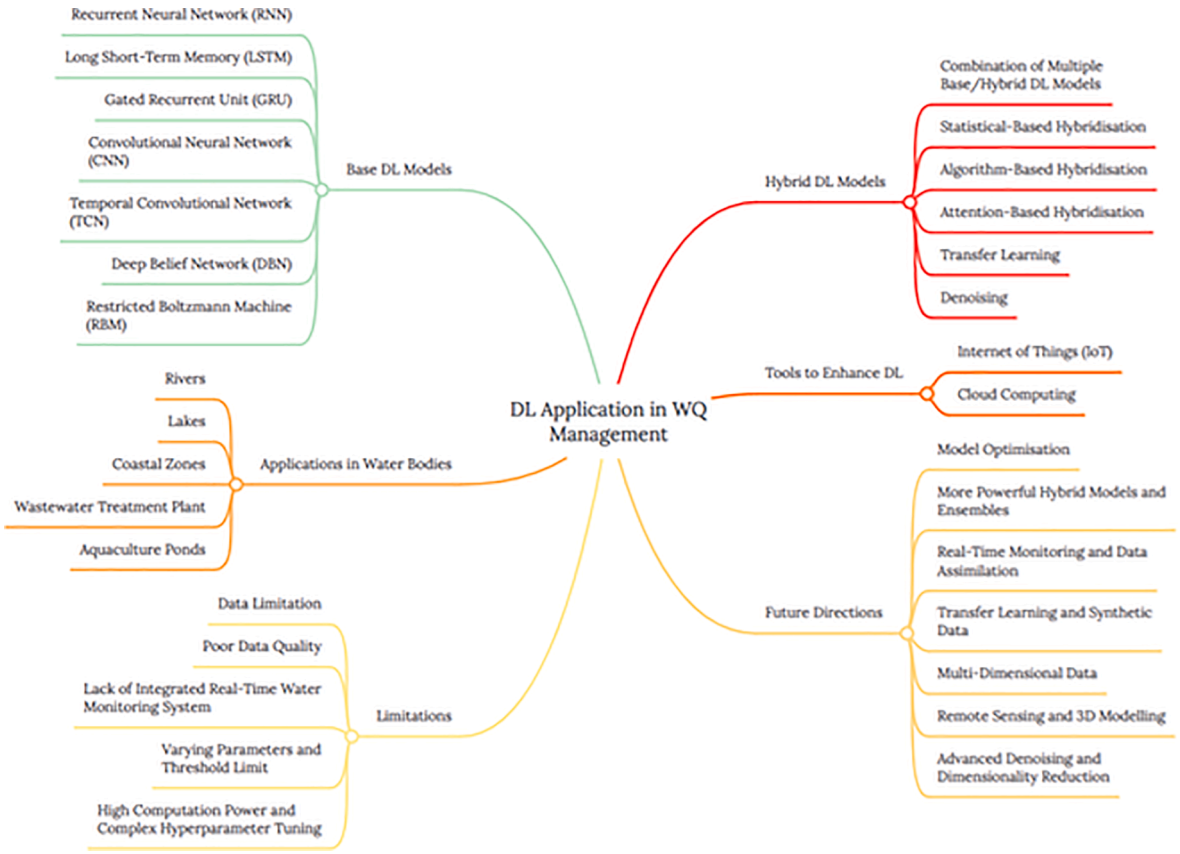


Fig. 2. Overall Review Framework.

treatment plants (WWTP). The AI tool is usually used for estimating the influent and effluent WQ parameters and controlling the operations of the WWTP to improve cost-efficiency. As such, the scope of the review remains relatively narrow and thus can be extended.

Tiyasha et al. (2020) conducted a literature survey of WQ modelling using AI models over the past two decades (i.e., 2000 to 2020). The review stressed various types of data input (in terms of the time scales and parameters) and output (WQ parameters), performance metrics used, and the variant of AI models. It should be noted that the papers reviewed by Tiyasha et al. (2020) only encompassed some well-known conventional AI models such as the artificial neural network (ANN), support vector machine (SVM) and the adaptive neuro-fuzzy inference system (ANFIS) models, as well as their hybrid counterparts (nature-inspired algorithms and wavelet transformation), but inadvertently having left out reviewing studies related to DL models. It must be stressed that the authors were only interested in the WQ of river water, which somehow or rather, made the review lack comprehensiveness.

In another similar review, Rajaee et al. (2020) reassessed the single and hybrid models in WQ management of river waters. The key difference between this review and that of Tiyasha et al. (2020) is that the technical focus of this review is on the techniques used for data preparation before the WQ modelling, which involves data splitting and data normalization. However, this review complemented the previous work of Tiyasha et al. (2020) by including some other models that are time-scales dependent such as the autoregressive integrated moving average (ARIMA) model and the Mann-Kendall analysis. Content-wise, the works of Tiyasha et al. (2020) and Rajaee et al. (2020) exhibited high similarities, and whereby extended works are needed in the future. In a recent study, Ighalo et al. (2021) enhanced the review scope to include all surface water bodies. Although the authors strived to include more water bodies in their review paper, the comprehensiveness of this publication was limited because the authors only reviewed ML or AI tools that had utilized the ANN model. Another literature survey by Mustafa et al. (2021) included the deployment of the internet of things (IoT) devices in a similar application. However, the paper only included an introduction to the IoT devices, without going into the substantial analysis and discussion on the importance of the subject matter.

Taking a closer look at Fig. 1, readers might observe that the share of publications related to the application of DL models in WQ management has been increasing since 2017. This can be attributed to the improvement in the computational facilities and the rejuvenation of DL models in the scientific communities. Yet, all the previous review papers did not include the role of DL models in WQ management in the scope of the study. Thus, a review paper that discusses the role and significance of DL models in the context of WQ management would indeed be most helpful and much needed. Hence, the authors of this review attempt to provide an inclusive review and literature survey on the application of DL

Table 1
Recommended Reading Priorities for Readers.

For readers who	Reading priorities (Sections)
Have an interest but do not have much background knowledge on DL	2, 3, 4, 6, 7
Are interested in using a recurrent-based neural network for WQ forecasting	2.1, 2.2, 2.3, 3.2, 4
Would like to know the common hybridization techniques and the latest applications of different DL algorithms on WQ forecasting	3, 4
Wish to develop signal decomposition-based hybrid models for WQ parameter forecasting	3.4
Are working on WQ research works that are relevant to IoT applications	5
Would like to identify the current limitations, future prospects and research gaps of WQ forecasting using DL methods	6, 7

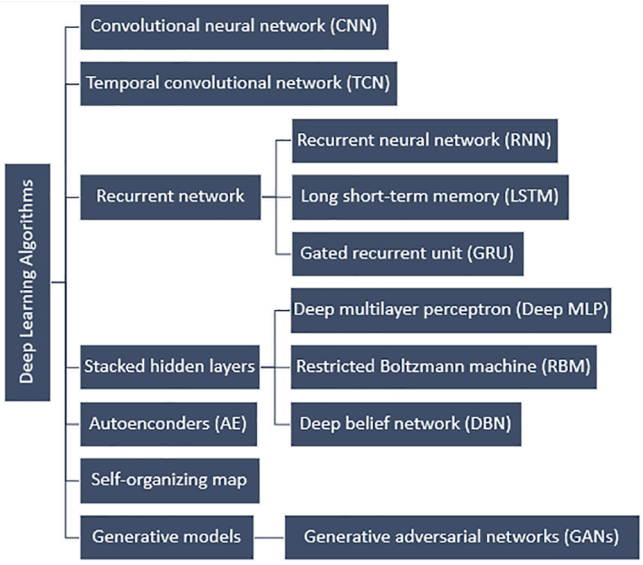


Fig. 3. Types of DL Algorithms.

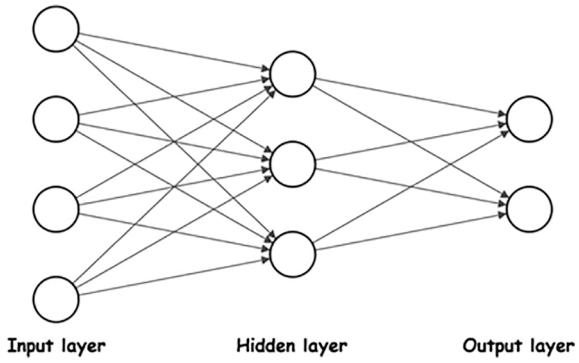


Fig. 4. The architecture of the ANN.

models in WQ management. The scope of this review paper shall not be constrained to the river water only, as portrayed by several earlier reviews, but also include every possible kind of water body. This review paper is expected to cover necessary elements such as background theories, modifications and applications of DL models in advanced WQ management.

The structure of this review paper is partitioned into several sections. Section 2 introduces the DL models that are commonly used for WQ management. Section 3 extends the discussion to include hybrid DL models via different hybridization techniques. In Section 4, the application of DL models in various WQ management scenarios is thoroughly reviewed. Section 5 provides an outlook for the IoT application concerning the DL model application in WQ management. Sections 6 and 7 give a visionary future trend and prospect of this research field before the concluding remarks. Fig. 2 illustrates the overall review framework which was created upon obtaining some meaningful inspiration from Herman et al. (2015). Table 1 is presented to ease the readers in locating the topics and information that they might be interested in. It summarizes the reading priorities recommendations for researchers and readers who may have different needs.

2. Deep learning (DL) models

DL, as the ML subdomain, comprises algorithms that build neural networks imitating the human brain for learning and analysis purposes (Dogo et al., 2019). It is described as an ANN approach with many layers of depth in the architecture, hence the associated phrase, 'deep'. Unlike the traditional shallow learning method, deep architectures model high-level abstractions in data via multiple non-linear transformations and enhance the extraction of in-depth information. Fig. 3 illustrates the types of DL architecture widely adopted for feature detection, clustering, dimensionality reduction, solving classification and regression problems. In the field of WQ management, the recurrent-based neural networks are widely adopted for WQ parameter forecasting.

2.1. Recurrent neural network (RNN)

2.1.1. Architecture of recurrent neural network (RNN)

The recurrent neural network (RNN) is an ANN that is specifically designed for handling sequential data. A typical ANN can be represented by a system of interconnected neurons as shown in Fig. 4, where the input layer receives the inputs, the hidden layers process the information and the output layer generates the outcome (Chia et al., 2020). The neurons in different layers are interconnected with some weights and run specific activation functions to map inputs to the output. In general, the neurons of the same layer should run identical activation functions; hence the type of the layer can be determined by the activation function itself (Sit et al., 2020). The multilayer perceptron (MLP), the most common form of the ANN, contains at least a hidden layer with each neuron connected to every neuron in the adjacent layer to form a fully-connected (FC) neural network. The ANN is also known as a feed-forward neural network because inputs were only processed in the forward direction. Owing to this, the ANN is unable to capture sequential information from the input data which is crucial for time series forecasting.

From Fig. 5, the forward propagation of the RNN is similar to that of the ANN with a single hidden layer but, the RNN receives an additional value calculated by the activation function at the previous time step. This looping constraint allows the RNN to generate memory states and build the relationship between data from different time steps. Despite being a strong learning model capable of handling temporal features, the RNN is difficult to be trained and suffers from exploding or vanishing gradient problems (Riedmiller, 1994; Chen et al., 2018). With the conventional back-propagation through time or real-time recurrent learning, the error signals backflow in time would lead to oscillating weights, rendering the learning process not working at all (Bengio et al.,

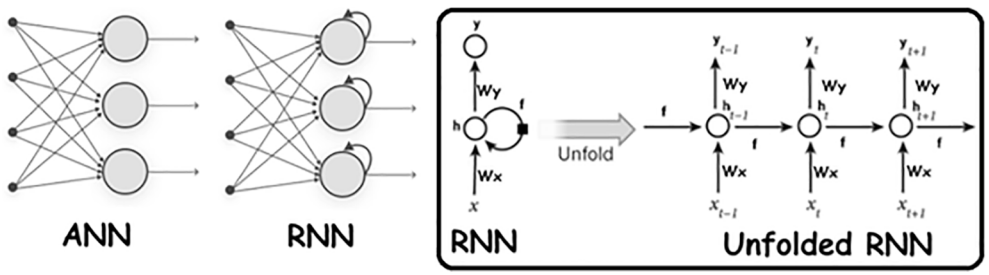


Fig. 5. Comparison of the ANN and RNN Networks.

1994; Werbos, 1988).

Echo state network (ESN) and simple recurrent unit (SRU). The base RNN is rarely used for WQ forecasting, but many researchers would compare their newly developed model with the base RNN to outline how much significant improvement has been achieved. Some variants were developed based on the RNN architecture, such as the echo state network (ESN) and the simple recurrent unit (SRU) (Liu et al., 2019a; Liu et al., 2020). The SRU simplifies the computation in such a way that each step is only dependent on the current input of the recurrence (Lei et al., 2017). This initiative helps to avoid the costly computation of matrix multiplication and hence the SRU could be trained as fast as a convolutional layer. When applied to water temperature and pH forecasting, the SRU demonstrated a higher degree of fitting than that of the RNN (Liu et al., 2019a). It was believed that the additional 'cell state' in the hidden layer of the SRU was responsible for judging and filtering effective information during the training process. Liu et al. (2020) further improved the SRU model by designing a bi-directional stacked SRU model whereby the forward and backward SRU were superimposed into each training sequence so that each point would have both the past and future context information. The proposed model was able to achieve $>$ 94 % of accuracy for multistep ahead forecasting and outperformed the RNN, LSTM and SRU. In comparison, the RNN was overfitting due to model complexity and did not converge to the global minimum.

Apart from these, a deep ESN was also developed for WQ prediction. The ESN model is an RNN with a hidden layer defining the state dynamics of the network (Jaeger, 2007). Kim et al. (2020b) designed a deep ESN model that is comprised of a hierarchy of stacked recurrent layers for which the output of each layer will be fed into the next layer. The proposed deep ESN model has demonstrated lower prediction errors than the extreme learning machine (ELM), random forest (RF), and gradient boosted regression trees (GBRT) in the prediction of BOD. However, the authors seldom compare their model performance with the benchmark LSTM and GRU models. As both the SRU and ESN were reported to have a lower computational cost along with better prediction performance, further studies focussing on the subject are highly recommended.

2.2. Long short-term memory (LSTM)

To solve the long-term dependency issues of the RNN, Hochreiter and Schmidhuber (1997) proposed the LSTM model with a gating mechanism. It is a type of recurrent neural network that has a cell state for remembering longer-term dependencies. Information usage and storage are modulated by a series of gates that can remove and add information to the cell state. A thorough explanation of the LSTM architecture can be found in Shen (2018) and Chen et al. (2018). In WQ parameter forecasting, the LSTM serves as a benchmark for comparison whenever researchers built a new model. The LSTM model also serves as the fundamental block for hybridization with other statistical tools, ML or DL models. For the DO and total phosphorus (TP) forecasting, the LSTM outperformed the back-propagation neural network (BPNN) and the online-sequential-ELM by demonstrating higher prediction accuracy and generalization regardless of data quality (Wang et al., 2017). Liang et al. (2020) showed that a simple LSTM structure with five hidden layers could simulate the forecast capacity of the Environment Fluid Dynamics Code (EFDC) in predicting chlorophyll-α concentration up to 31 days ahead. Apart from these, the LSTM was used to process remote sensing-based datasets and make predictions on coastal algae bloom (Shin et al., 2019).

The LSTM has been frequently modified by researchers and now comes in several variants. Graves and Schmidhuber (2005) first proposed the bi-directional LSTM (Bi-LSTM), whereby the data was fed in both the normal and reversed time order, while there will be no communication between both hidden layers to ensure that the expanded graph is acyclic. Some researchers reported that the Bi-LSTM learns faster than the unidirectional approach in handling sequential data (Siami-Namini et al., 2019; Abduljabbar et al., 2021). For wastewater COD prediction, the Bi-LSTM outperformed the ML models (SVM and regression tree) as the former could control data propagation to prevent gradient explosion (Guang et al., 2019). However, the standard LSTM fared relatively well than the Bi-LSTM since the COD prediction is a one-sided problem. This finding is in contradiction to that of Cheng et al. (2020) who found that the Bi-LSTM outperformed the LSTM in WWTP key features forecasting. The authors believed that the Bi-LSTM employed both historical and future data to improve its prediction performance. Since a valid conclusion could not be drawn from these studies, further studies on the above subject are recommended to challenge the models in more WQ datasets.

Cho and Park (2019) designed a merged-LSTM that combined the outputs from three parallel LSTMs into a fully-connected layer for multi-step ahead chlorophyll-α prediction. Unlike a conventional LSTM model, additional WQ and weather data from a nearby station were fed into different LSTMs and trained separately. The merged-LSTM performed better than the conventional LSTM and MLP models but at a higher computational time cost. Nevertheless, Song et al. (2020b) used the merged-LSTM for sea surface height anomaly detection. This is a time series forecasting problem where the hidden discipline in short time series was extracted to tackle the long-term dependence of series changes. The application of the merged-LSTM could be extended to the spatiotemporal prediction of WQ in lakes, mariculture ponds, and WWTP, based on auxiliary data collected from nearby stations. The correlation analysis or dimensionality reduction tools may be handy in terms of reducing the complexity of the auxiliary data. Other recurrent-based networks could also be merged for better performance in the spatiotemporal prediction of WQ parameters.

2.3. Gated recurrent unit (GRU)

Cho et al. (2014) introduced a gated recurrent unit (GRU) network that modulates information flow without using separate storage, where only two gates are present in the base GRU. The reset gate is responsible for the short-term memory by combining new input and historical memory, which is analogous to the combination of input and forgot gates in the LSTM. The update gate, on the other hand, behaves like an output gate and decides how much of the previous state shall be kept for long-term memory. A detailed explanation on the GRU model architecture and its structure is given in Chen et al. (2018). In general, the GRU exhibited better performance than the LSTM on a small dataset (Gruber and Jockisch, 2020). The GRU also converged much faster than the LSTM with a higher training efficiency (Cheng et al., 2020). Qin (2019) reported that the GRU outperformed the convolutional neural network (CNN), LSTM, temporal convolutional network (TCN) and the Bi-directional RNN (Bi-RNN) for forecasting DO time series. However, on the contrary, Loc et al. (2020) showed that the GRU was sensitive to the extreme values in the dataset, while the LSTM was more successful in representing a time series with smooth and repeating patterns without random fluctuations. When dealing with a large dataset, the GRU may fail to capture a long-term correlation and be trapped in a gradient vanishing problem (Song et al., 2020a). Thus, the GRU provides a trade-off between accuracy and training time for practical problems involving a large dataset with a long temporal sequence (Zhang et al., 2018a). It is worth mentioning that both the LSTM and the GRU often outperformed the RNN and other ML models, given that the former can preserve important features through various gates to ensure they will not be lost during long-term propagation. As a result, both the GRU and LSTM currently represent the state-of-the-art and most highly benchmarked modern models in many applications.

Table 2
Comparisons of the DL Models.

Model	Architecture	Characteristics	References
RNN	• The simplest recurrent network that maps input and output vectors for the entire historical dataset	• Easily trapped in vanishing or exploding gradient problem • Not suitable for handling large WQ datasets with long temporal sequence	(Riedmiller, 1994; Chen et al., 2018)
LSTM	• A recurrent-based network with cells comprised of an input gate (choose what to store), forget gate (choose what to discard) and output gate (choose what to pass to the next hidden state)	• Suitable for long temporal prediction with high accuracy at a higher computational cost • Conquers vanishing and exploding gradient problem • Suitable for WQ time series forecasting involving a large WQ dataset with long temporal sequence	(Chen et al., 2018; Zhou, 2020; Loc et al., 2020)
GRU	• A recurrent-based network based on the LSTM, but reduced to only two gates in the GRU, namely the update gate (decide which information to be discarded and which new information to be added) and reset gate (decide how much past information to forget)	• Moderately high accuracy but has a substantially lower computational cost than LSTM • Often perform better than LSTM in a small dataset • Possible to have gradient vanishing problem for very long-term correlation • Suitable for WQ time series forecasting involving a moderately large WQ dataset	(Yan et al., 2021; Li et al., 2021c; Gruber and Jockisch, 2020)
CNN	• Comprised of a convolution layer (filter of predefined size to learn features), pooling layer (to reduce the size of the input), and FC layer (for generating output) • The convolutional layer is responsible for the extraction of spatial features	• Exhibits poor performance in capturing temporal features but is an excellent feature extraction tool in handling high dimensional data • Suitable for WQ studies involving multi-dimensional datasets, remote-sensing images and classification tasks	(Wang et al., 2020a; Fan and Zhang, 2020; Xue et al., 2021; Habib and Qureshi, 2020).
TCN	• Based on the architecture of CNN and RNN, the TCN uses dilated convolutions and residual blocks to enhance the utilization of long historical observations without a vastly deep structure	• Dilated convolution allows the network to capture local and temporal information • Handle long input sequences by using less memory • Prevent exploding and vanishing gradient problem	(Zhang et al., 2019; Qin, 2019)
RBM	• A shallow two layers neural network with all the nodes on visible layers connected to every node on the hidden layer • The RBM reconstructs individual weights and overall bias in an unsupervised manner	• Shows vanishing gradient problem and is not suitable for data with long temporal dependency • Few parameters to be tuned • Automatically detects the significant relationship between input features	(Hinton et al., 2006; Zhang et al., 2018b; Wu et al., 2015)
DBN	• Comprised of the stack of RBMs with bipartite connections removed, except for the top two layers. • The model is trained by a Greedy Layer-wise algorithm to allow each layer to learn the full feature of the whole dataset	• Contain more parameters to be optimized along with a higher capacity per parameter • Fairly efficient training procedure whereby unsupervised learning for feature detection and supervised learning to fine-tune the features to optimize discrimination • Data need not be labelled during pre-training to minimize human errors	(Ren et al., 2020; Qin et al., 2017; Niu et al., 2020; Kuremoto et al., 2014)

2.4. Convolutional neural network (CNN)

A typical CNN is comprised of the convolutional, pooling and FC layers, and adopts the way of local connection and weight sharing. The convolution layer contains a series of filters that slide across the image to make a dot product with the input data, the pooling layer reduces the dimension of the matrix produced, and the FC layer condenses the extracted feature to produce the final result (Hu et al., 2020a). For a more extensive explanation of the CNN architecture and computational approaches, readers could refer to a state-of-art review article published by Alzubaidi et al. (2021).

As the CNN is capable of extracting sophisticated features from multidimensional data, it can be used to process imagery data for regression and classification. For instance, Song and Kim (2020) employed the CNN for building pollutants load prediction model using hydrological images as the input. The CNN was also trained on spatio-temporal WQ data generated by a hydrological model for the algal bloom prediction (Pyo et al., 2020). The model maintained a satisfying performance with a significant decrease in the temporal frequency and the volume of spatial data. Xue et al. (2021) reported that the CNN outperformed the traditional models such as the linear regression (LR) and partial least squares regression (PLSR), in estimating lake chlorophyll-α based on remote sensing images. Apart from the prediction accuracy, the weight sharing characteristic of the CNN decreases the number of trainable network parameters and speeds up the training process. The neurons corresponding to the same features will be triggered in varied scenarios, hence would improve the generalization ability of the model. Also, training a model on low-resolution data saves time and allows for high-resolution data to be predicted more quickly (Alzubaidi et al., 2021). Owing to this, researchers hybridized the CNN with other time series forecasting models to extract important features from a large dataset. Such a hybridization network would significantly lower computational time cost and improve model prediction performance, and this is discussed in the next section.

2.5. Temporal convolutional network (TCN)

The temporal convolutional network (TCN) was proposed by Bai et al. (2018) for sequence modelling tasks. The TCN adopts the 1D fully convolutional network architecture so that both input and output would share the same length. The building block of TCN, which is commonly known as the residual unit, stacks two dilated causal convolution layers together where the causal convolutions ensure that future information does not leak into the past (Qin, 2019). By skipping the connection within the residual unit, the TCN allows propagation with a larger but stable gradient along the neural network. However, the TCN takes in the raw sequence up to the effective history length during evaluation and may perform poorly during domain transfer due to a small receptive field. The TCN has demonstrated excellent performance in anomaly detection owing to its capability in exploring the temporal dependencies over a remarkably long historical period (Song et al., 2020a). Owing to the same reason, the TCN also exhibited better performance than the recurrent counterparts in predicting the Brunett River dissolved oxygen (DO) and water temperature (Zhang et al., 2019). It was believed that the TCN adopted convolutional filters of different sizes to collect temporal dependencies at different time scales. With the dilated convolutions that increase the receptive field, the long historical data can be utilized for learning and prediction purposes. However, Qin (2019)

reported that the GRU outperformed the TCN in predicting DO at the Yangtze River. Based on the study, the GRU is probably a better option over the TCN, for WQ prediction, despite of the plenty factors that may affect the performance of the models. As such, more studies can be conducted to challenge the TCN in different types of WQ datasets before a valid conclusion is drawn.

2.6. Deep belief network (DBN) and restricted Boltzmann Machine (RBM)

The restricted Boltzmann machine (RBM) comprised of a visible and a hidden layer, which can automatically detect inherent patterns in the data by reconstructing the inputs. However, the RBM is unable to tackle vanishing gradient problems, hence using the deep belief network (DBN) would be more favourable in most situations. The DBN is designed in such a way that multiple RBMs are stacked together, where the output of a low-level RBM would serve as the input for the high-level RBM (Wu et al., 2015). The training process is repeated for all layers to ensure each RBM learns the entire input; unlike in the CNN where the convolutional layer detects patterns and the pooling layer recombines them subsequently. A detailed explanation of the architecture of the DBN can be found in Shen (2018). The DBN has been widely adopted for many applications including WQ point estimation and time series forecasting. Most WQ-related studies employed a five-layered DBN with three stacked RBMs as hidden layers (Zhang et al., 2016; Lin et al., 2018; Ren et al., 2020). The first RBM receives the inputs while the high-level RBMs further extract the characteristic of influencing factors with their weights updated automatically. The deep architecture of the DBN with adequate hidden layers has demonstrated better generalization and accuracy than the BPNN (Zhang et al., 2016). Similarly, Ren et al. (2020) found that the DBN outperformed the decision tree (DT) and CNN regression models in aquaculture pond DO prediction. However, the authors did not compare their models with the LSTM and GRU, despite both being well-known for handling temporal sequence data. The genetic algorithm (GA) was used to optimize the DBN for effluent COD and SS prediction (Niu et al., 2020). The study showed that the optimal number of RBM layers varies according to different datasets and applications and scholars may refer to the methods proposed by Montúfar (2014) to approximate the minimum number of layers required for their DBN network. Apart from modifying the internal structure, the DBN applications in WQ-related studies involving hybridization with other models and this will be discussed in Section 3.

2.7. Comparisons of deep learning models

Table 2 summarizes the architecture and characteristics of each of the DL models. This could serve as a quick reference for future scholars who wish to have a fresh start in selecting an appropriate DL model in the WQ prediction. As a rule of thumb, the researchers shall first adopt the LSTM and the GRU for conventional WQ time series forecasting.

3. Deep learning hybrid models

3.1. Convolutional neural network (CNN) hybrid models

Hybridization of models of different architectures may allow inherent information about the data to be extracted. Given that the CNN is specifically designed to handle data with similar grid weight, the characteristic of local connection, weight sharing and pooling could effectively reduce network complexity and training parameters. For instance, Fan and Zhang (2020) reported that the CNN outperformed the principal component analysis (PCA) in extracting useful features when coupling with the SVR for BOD prediction. In another study, Lee et al. (2020b) opined that each layer of the CNN takes a few adjacent features and may not fully consider relationships among all input features. A higher-order partial least squares (HOPLS) regression model was hybridized with the CNN to improve its performance in algal bloom prediction. Despite the CNN could effectively extract key input features from multi-dimensional datasets, the standalone CNN gave the worst performance in predicting ammonia as compared to the GRU, LSTM and SVR (Wang et al., 2020a). Similar findings were reported by Wang et al. (2019b) and Barzegar et al. (2020) who developed the CNN-LSTM for the WWTP and lake WQ prediction, respectively. In the studies, the CNN serves as an ideal feature extractor when hybridized with the time-series forecasting model. Apart from improving prediction accuracy, the CNN reduces the complexity of the training parameters and converges with lesser epochs (Balderas et al., 2019).

3.2. LSTM-based and GRU-based hybrid models

3.2.1. Hybridization with convolutional neural network (CNN)

Wang et al. (2016) first designed the CNN-LSTM to predict valence-arousal ratings. The regional CNN divides input text into several regions so that useful information in each region could be extracted and weighted, then sequentially integrated via the LSTM for valence-arousal rating prediction. Similarly, the CNN-LSTM was adopted for handling multi-dimensional WQ, where the CNN extracts deep features of the main elements while the LSTM performs temporal forecasting based on these in-depth features. The CNN-LSTM has been reported with higher stability when handling data with large fluctuations (Wang et al., 2019b; Baek et al., 2020). In another study, Barzegar et al. (2020) discovered that the LSTM was better in predicting the high concentration of parameters while the CNN was good at low concentration situations. The hybrid CNN-LSTM model combined the strengths of both models and enhanced the prediction accuracy (Baek et al., 2020).

The GRU, which regularly outperforms the LSTM, was also hybridized with the CNN for WQ prediction (Tu et al., 2019; Hu et al., 2020a; Song et al., 2020a). The strength possessed by the CNN and GRU in processing high-dimensional and time sequence data, respectively, renders both models a perfect match for handling spatiotemporal datasets. The architecture of the CNN-GRU is nearly comparable to that of the CNN-LSTM, except for the recurrent module. Apart from the superior stability, the CNN-GRU could be trained faster as the CNN had sped up the process of inherent feature extraction (Tu et al., 2019; Pan et al., 2020). In another study, Song et al. (2020a) used the CNN as a down-sampling network and assisted the GRUs in extracting the overall trend and the local short-term pattern from the dataset. As a result, the CNN-GRU outperformed the statistical ARIMA, ML (SVR and ANN) and DL (RNN and TCN) models in predicting sea surface salinity (SSS) 14 days ahead.

3.2.2. Hybridization with statistical models

Traditional statistical models use the linear relationship between data to capture its trend and face some shortcomings when dealing with complex non-linear relationships between features. On the other hand, the DL models offered better predictions but a massive amount of training data would be required. As such, Xu et al. (2019) coupled the seasonal autoregressive integrated moving average (SARIMA) and the LSTM for coastal pollutant prediction. The authors first reported that the SARIMA could not react to the sudden change while the LSTM failed to capture annual periodicity, probably due to the limited amount of training data provided. The SARIMA-LSTM hybrid model was improved based on these two models and resulted in a lower prediction bias. By providing auxiliary data to the multi-featured SARIMA-LSTM, the model prediction accuracy and stability were further enhanced. To the best of the authors' knowledge, the hybridization of the statistical model and the GRU for WQ modelling is rarely reported, thus could be an avenue for future study.

3.2.3. Hybridization with other ML algorithms

As an unsupervised learning neural network, the autoencoder (AE) had been widely used in data compression, dimensionality reduction

and feature extraction projects, for decades. A conventional AE comprises an encoder that maps the inputs into the code, while a decoder performs a decoding task to reconstruct the inputs. For a detailed explanation of the algorithm, the readers are referred to Zhang and Jin (2020). In terms of application, the AE was employed for filling missing data in the WQ time series (Shu et al., 2021). Li et al. (2018) adopted the sparse autoencoder (SAE) to extract deep latent features from WQ and meteorological data, and significantly improved the performance of the LSTM. However, the spatiotemporal correlation of the WQ fluctuations was not considered; hence there is still room for improvement. Since then, Zhang and Jin (2020) proposed the AE-LSTM for spatiotemporal prediction of river nutrients which had outperformed the conventional LSTM. The excellent performance of this hybrid model can be attributed to (i) establishing spatiotemporal characteristics of the WQ parameters in the upstream and downstream stations (ii) feature dimensionality reduction using the AE (iii) superiority of the LSTM algorithm for processing time series.

The gradient boosting decision tree (GBDT) was also coupled with the LSTM for aquaculture DO forecasting (Huan et al., 2020). The GBDT effectively performed feature selection to reduce unnecessary information in the model thus resulting in higher prediction accuracy. It is surmised here that the proposed GBDT-LSTM models could be gainfully challenged with the GRU, a competing model of the LSTM. As such, the performance of the CNN, AE and GBDT as feature extraction tools could be compared accordingly. To handle a sparse dataset, the factorization machine with the seq2seq framework was coupled with the GRU for DO forecasting (Xu et al., 2021). The factorization machine obtained interacting correlations between diverse index values and processed them as an input to the GRU model, thus significantly improving model prediction accuracy and outperforming the conventional DL, ML and statistical models. In another study, the gradient boosting regression tree (GBRT) was coupled with the attention-based GRU for three-dimensional (3D) DO forecasting, where the GBRT was used as a spatial prediction model to estimate local DO concentration in the pond (Cao et al., 2021). The GBRT is capable of processing a wide range of data types and is quite resilient when dealing with outlier data points, thus granting the model additional advantages when dealing with a small dataset of fewer features.

3.3. Deep belief network (DBN) hybrid models

Given that improper design would lead to longer training time and poor accuracy, Qiao et al. (2018) proposed a self-organizing DBN network to adjust the number of hidden layers and hidden neurons based on an automatic growing and pruning algorithm. The self-organising DBN has accelerated the learning process and demonstrated robustness in tolerating input fluctuations. Wang et al. (2020b) proposed the recursive-time DBN in such a way that the input layer received the current moments and historical time values of the characterization factors and influencing factors. The resulted hybrid model outperformed the base DBN and LSTM models in forecasting algae density. To improve the model training efficiency, Wang et al. (2021) proposed the DBN with an event-triggered learning algorithm (DBN-EL) to ignore ineffective data pairs such as outliers. The DBN-EL converged much faster without noticeable fluctuations as it benefited from the selective response mechanism. Nevertheless, the prediction accuracy of the DBN-EL was higher than the conventional DBN and sparse deep belief fuzzy neural network by demonstrating higher fitting. Further experiments also showed that the DBN-EL has a stronger capability in predicting TP concentration with a shorter runtime.

Apart from using the DBN as a forecasting model, Qin et al. (2017) used the DBN as an unsupervised feature extractor to assist the ARIMA in red tide biomass forecasting. Wang et al. (2020c) coupled the DBN with multivariate space–time autocorrelation (STVAR) to extract useful features from the multivariate WQ and weather data, then fused it into the ANFIS for spatiotemporal chlorophyll-α prediction. In another study, the DBN excavated essential characteristics of the WQ parameters for total nitrogen (TN) prediction and effectively reduced overall prediction errors by 50 % as compared to that of standalone LSSVR (Yan et al., 2020).

3.4. Denoising-based hybrid models

Denoising algorithms such as the empirical mode decomposition (EMD), ensemble EMD (EEMD) and the CEEMDAN, are frequently used to decompose original time series data into several stable intrinsic mode functions (IMFs) to ease modelling. In some cases, the high-frequency IMFs may be recognized as noise and are to be removed from the model input. The EMD is a signal adaptive decomposition technique proposed by Huang et al. (1998) that decomposes complicated signals according to the local characteristic time scale. The EEMD which was proposed by Wu and Huang (2009) is mainly used to overcome the mode mixing problem in the EMD through the addition of an ensemble of white noise to the signal. The added white noise generates a uniform reference frame in the time–frequency space so that the signal of a similar scale can be collated into the same IMF. Torres et al. (2011) then invented the CEEMDAN that removes residue white noise completely upon decomposition, which ensured a better spectral separation of modes at lower computational cost. The mathematical theories behind the denoising algorithms could be referred to in the said literature.

The excellent performance of denoising algorithms in improving the ML models has been reported by Lu and Ma (2020). The authors also suggested that choosing an appropriate signal denoising method is more crucial than the model since the data quality is the essence of modelling. Eze and Ajmal (2020) decomposed DO time series data and selected the highly correlated IMFs for the LSTM training. The EEMD-LSTM was able to follow the trend of DO closely up to 12 h and outperformed the SAE-LSTM proposed by Li et al. (2018). Zhang and Yang (2020) reported that the CEEMDAN-GRU has consistently captured the trend of suspended sediment concentration in the Yellow River and had outperformed the GRU without a denoising algorithm. Huang et al. (2021) further improved the performance of the CEEMDAN through the addition of the Lempel-Ziv complexity, which is a symbolic dynamic algorithm used for quantifying the uncertainty in time series data (Lempel and Ziv, 1976). The IMFs with similar Lempel-Ziv complexity values were stacked for the GRU training. The proposed model demonstrated lower prediction errors and higher fitting than the CEEMDAN-GRU. A similar result was observed when the GRU was replaced with the LSTM and RNN.

Dragomiretskiy and Zosso (2014) developed the variational mode decomposition (VMD) that non-recursively decomposes an input signal into IMFs comprised of amplitude and frequency modulated waves. With the aid of the Wiener filter, the VMD is not prone to background noise, thus outperforming the EMD in many applications (Maji and Pal, 2016; Uzunoglu, 2018). Ren et al. (2020) reported that the VMD significantly reduced the noise and highlighted key data features for better DO prediction in aquaculture ponds. In another study, Fijani et al. (2019) proposed a two-staged decomposition method. Upon the CEEMDAN decomposition, the high-frequency IMF was further resolved into finer-scale components by the VMD so that the model could be more responsive to the fine-scale data patterns. The CEEMDAN-VMD-based models were validated more accurately than the CEEMDAN-based models. Similar findings were also reported by Dong and Zhang (2021) where the CEEMDAN-VMD-based models were better in extracting multiscale periodicity and non-linearity of the pollutant time series. Hence, the two-staged decomposition involving different signal denoising techniques could be attempted whenever a single decomposition technique did not perform satisfactorily and a comparison between different denoising algorithms on model performance could be studied.

3.5. Attention-based hybrid models

The attention algorithm was proposed to improve model

performance in machine translation by attributing the highest weights to the most relevant parts of the input sequence (Zhu et al., 2019). A detailed illustration of the mechanism is given in the literature (Bahdanau et al., 2014). In WQ prediction, the weather or WQ parameters could be some influencing factors, hence the attention mechanism may select key input features for target prediction by assigning larger weight to the essential parameters. Bi et al. (2020) incorporated an attention layer into the LSTM for effective information extraction from the long temporal dependence DO dataset. Zhu et al. (2021) designed a hybrid model comprised of the ResNet, Bi-LSTM and multi-head attention to extract features from different WQ time series, then concatenated into FC layers for output prediction. In another study, the CNN-LSTM coupled with a self-attention mechanism has led to a better global convergence in predicting the effluent quality of WWTP (Li et al., 2021c). The author explained that the CNN-LSTM model was driven by the attention mechanism and thus can easily stay focused on the salient features in the dataset.

For 3D WQ modelling in an aquaculture system, the attention mechanism established long-distance dependence between input sequences and highlighted the impact of key features on DO prediction (Cao et al., 2021). Liu et al. (2019c) compared different types of attention mechanisms and reported that the temporal and spatiotemporal joint attention mechanisms outperformed others in DO forecasting. The temporal attention-based RNNs can maintain higher accuracy over a long-term DO prediction and are highly recommended for the WQ time series prediction.

3.6. Transfer learning-based hybrid models

Being different from the traditional ML techniques where the model tries to learn everything from scratch, the transfer learning (TL)-based model typically transfers knowledge from the previous task to the target task where the latter has less training data (Pan and Yang, 2010). By applying the knowledge learned from the source domain, the TL algorithm would improve generalization in the target domain and avoid overfitting (Yosinski et al., 2014). Knowledge transfer can be categorized into a data distribution transfer, domain transfer and task transfer and most studies involve domain transfer (Lumini and Nanni, 2019; Zhou, 2020; Zhu et al., 2021). Wang et al. (2019a) designed a growing DBN network based on TL. A single layer DBN was pre-trained as the source domain, transferred to the target domain to initialize its weights and allowed the DBN to grow in depth. When applied to the actual dataset, the TL-DBN demonstrated higher learning speed and was more robust than the existing structure. Tian et al. (2019) applied TL on the ANN, RNN and LSTM for chlorophyll-α dynamic prediction. The models were pre-trained using WQ data collected in June, then fine-tuned and applied for chlorophyll-α prediction based on test data collected three months later. The TL-based models have reduced the influence of mutable data distribution and maintained higher accuracy for prediction in the next three months. The authors also reported that the TL was better than the parameter norm penalties and dropout methods in improving model generalization ability.

The TL was applied to WQ classification based on remote sensing images (Pu et al., 2019). *In situ* data were first collected from a lake in China and calibrated with Landsat-8 images as a pre-training for the TL-CNN. The model was then applied to another lake lacking *in situ* measurement data. The TL-CNN model outperformed the standard CNN, SVM and RF along with satisfying generalization ability, and thus it was recommended for inland-lake monitoring where *in situ* data collections are rather difficult. In another study, the TL-LSTM was used for river DO forecasting under severe data missing scenarios where direct training of the model is not feasible (Zhou, 2020). The LSTM was first trained with complete data to learn its pattern, then transferred to the target temporal sequence with missing data for DO forecasting. As a result, the TL-LSTM could follow the DO trend adequately under different data missing scenarios, unlike the LSTM model that demonstrated apparent flatten prediction. Similarly, Zhu et al. (2021) pre-trained their DO prediction model at Taihu Lake using a large dataset and subsequently applied the model at Yixing Lake with limited data. It was found that the TL-based model had higher prediction accuracy and shorter training duration than the model that was only trained with data collected from Lake Yixing. Despite the differing water characteristic at both lakes, the TL was proven effective and significantly reduced the amount of data required. In most situations, the TL algorithm was paired with the DL to solve the data limitation and reduce training time in the target domain. Since only a small amount of labelled data is available, model training is rather difficult and tends to overfit. By having the model pre-trained in a larger dataset, it could initialize model weights for the target domain and hence contributes to better performance than starting from scratch. This would avoid random initialization of parameter weights and increase model training efficiency typically in a deep architecture (Qureshi et al., 2017). As such, the TL algorithm is highly recommended for WQ prediction as data availability is always a major issue.

3.7. Process-based hybrid models

The process-based models are mechanistic models that work on general physical relationships such as mass and energy conservation to simulate the target variables. As these models are strongly rooted in scientific theory, they were designed with a rigid relationship with several constraints and often diverged from the theory when calibrated with real-world data (Read et al., 2019). When applied in the field of hydrology, the auxiliary parameters like climate or topography data would increase the difficulties of model calibration. On the other hand, the DL models have no assumptions of processes underlying the data. It might be difficult in several ways for prediction in unmonitored regions or long-term projection due to data sparseness, limited auxiliary data (weather or WQ data from nearby stations) and the impact of some expected climate changes (Varadharajan et al., 2022). A large dataset would also be required for the DL to learn the dynamics of the complex systems and to make accurate predictions.

In recent years, the idea of integrating scientific knowledge and the RNN-based models gained momentum and growing interest amongst researchers (Willard et al., 2020). The knowledge-guided DL hybrid models were developed for downscaling, inverse modelling, creating synthetic data, discovering governing equations, simplifying the complex mechanistic model and parameterization. This parameterization is not the commonly defined terminology found in the mathematics field but refers to the process of replacing a complex mechanistic model with a simplified one so that it can be modelled using DL strategies. Some process-based hybrid models attempt to make physically consistent and robust predictions by designing a loss function to penalize the model when a known physical law or relationship is violated (Read et al., 2019).

A process-guided (PG) recurrent graph network was proposed for modelling interacting segments in the river network (Jia et al., 2020). The prior knowledge embedded in the physical model was used to improve the learning of latent representation while the loss function was optimized to improve global and local prediction on each river segment. Sadler et al. (2022), in turn, proposed a PG-based LSTM to forecast streamflow and temperature simultaneously via a multi-task learning approach. Given that streamflow and temperature formed part of the energy transfer processes, the model established their relationship for multi-objective optimization and showed significant improvement over the single-task model that predicted both parameters separately. From these studies, the hybridization of recurrent-based models with process-based models has shown promising results in different applications. It remains as a great research direction to solve to complex science and engineering problems by integrating traditional physics-based modeling approaches with different state-of-the-art DL models.

Table 3
Past Research on River WQ Prediction using DL Algorithms.

Authors	Parameter	Study Area	Model	Input Data	Research Remarks
(Dong et al., 2019)	DO	GuBeiKou River Basin, China	SG-ED-LSTM LSTM	DO at 4 h interval 10,000 sets	• The LSTM outperformed ANN and ARIMA. • Encoder-decoder (ED)-LSTM with Savitzky-Golay (SG) filter converged faster with lower MAPE in 5-steps ahead DO forecasting
(Qin, 2019)	DO	Yangtze River, China	GRU	DO at 30 min interval 64,411 sets *(2012 to 2016)*	• The GRU outperformed CNN, TCN, Bi-RNN and LSTM. • The prediction capability of all models dropped drastically when forecasting horizon > 5 steps.
(Zhang et al., 2019)	DO, temperature	Burnett River, Australia	MTCN TCN	Temperature, EC, pH, DO, turbidity and chlorophyll-α at 30 min interval	• Both TCN and multi-task learning TCN (MTCN) accurately captured the trends of both parameters but the MTCN took half of the training time required by TCN.
(Bi et al., 2020)	DO	GuBeiKou River Basin, China	SG-Attention-LSTM LSTM	DO at 8 h interval 10,000 sets *(2014 to 2018)*	• SG filter showed a better filtering effect • The SG-Attention-LSTM showed higher accuracy than LSTM, seq2seq, XGBoost, SVR, ANN and ARIMA.
(Zhang et al., 2020)	DO	Burnett River, Australia	PCA-RNN	Temperature, EC, pH, DO, turbidity and chlorophyll-α at hourly interval 3672 sets *(Jun to Oct 2015)*	• The DO trend followed a similar daily pattern while the turbidity and chlorophyll-α showed fluctuations due to weather events and river discharge. • The PCA-RNN outperformed ANN and SVR in 1, 2 and 3 h DO forecasting.
(Zhou, 2020)	DO	River in Shanghai City Island, China	TL-LSTM LSTM	DO, AN, COD, pH, ORP, EC, turbidity, temperature and water level at hourly interval 3,157,920 sets *(2015 to 2018)*	• The TL-LSTM outperformed the LSTM in 2, 6 and 10 h DO prediction under different data missing scenarios. The TL-LSTM followed the DO trend adequately whereas the LSTM showed a flattened prediction.
(Xu et al., 2021)	DO	Lianjiang River, China	FM-GRU GRU LSTM	pH, EC, DO, temperature and turbidity at hourly interval 13,128 sets *(2019 to 2020)*	• Factorization machines (FM) were integrated with the seq2seq framework to solve missing data problems, and the proposed FM-GRU has out-performed GRU and LSTM.
(Kim et al., 2020b)	BOD	Han River, South Korea	ESN	BOD, pH, EC, DO, temp, COD, SS, TN and TP 1000 sets *(2008 to 2018)*	• The authors estimated BOD based on different combinations of WQ inputs. The ESN model performed well without SS, TN and TP, thus outperforming the ELM, GBRT and RF model.
(Khullar and Singh, 2021)	BOD and COD	Yamuna River, India	CNN-Bi-LSTM CNN-LSTM LSTM	DO, pH, free ammonia, COD, BOD, EC, temperature and coliform at monthly interval 3240 sets *(2013 to 2019)*	• The hybridization of CNN and Bi-LSTM reduced model complexity, and the resulted CNN-Bi-LSTM outperformed CNN-LSTM, LSTM, LR, RF, ANN and SVR in forecasting river pollutants.
(Tu et al., 2019)	COD	Shanghai Jinze River, China	CNN-GRU GRU RNN	COD at 1 min interval 300,520 sets *(2017)*	• The CNN-GRU outperformed the GRU, RNN and SVR by demonstrating lower errors and shorter training duration.
(Hu et al., 2020a)	COD	Xihe River, China	CNN-GRU	COD, water flow and flow velocity at hourly interval *(Jan 2019 to Apr 2020)*	• KNN was used to remove outliers, and a 3-layered GRU activated by ReLu performed the best. The CNN-GRU showed higher prediction accuracy than the GRU, BPNN and ARIMA.
(Lee and Lee, 2018)	Algae	Han, Geum, Yeoungsan and Nakdong River, South Korea	LSTM RNN	Temperature, pH, DO, BOD, COD, chlorophyll-α and cyanobacteria at weekly interval. Water level and pondage at daily interval 4464 sets *(2012 to 2017)*	• The LSTM was capable of solving the problem of information loss and was superior to MLP and RNN. • The LSTM, RNN and MLP performed better than an analytical model in forecasting chlorophyll-α.
(Cho and Park, 2019)	Algae	Geum River, South Korea	Merged-LSTM LSTM	Chlorophyll-α, TN, TP, TOC, DO, EC, temperature, pH and weather data at daily interval 981 sets *(2013 to 2018)*	• A merged-LSTM was developed by combining 3 parallel LSTMs to handle auxiliary data and weather data from nearby stations. The model was trained slower but was more accurate than the LSTM and MLP.
(Wang et al., 2020c)	Algae	Water monitoring station in Jiangsu, China	DBN-STVAR-ANFIS DBN	Chlorophyll-α, DO, TN, temperature, air temperature and wind speed 1095 sets over 3 years	• The DBN and STVAR extracted features and fused them into ANFIS for spatiotemporal chlorophyll-α prediction. • The proposed DBN-STVAR-ANFIS had lower RMSE than the DBN and time autocorrelation models.
(Jang et al., 2020)	Algae	Geum River, South Korea	LSTM-EDFC	Temperature, pH, EC, DO, TOC, TN, TP, nitrate, AN and weather data at monthly interval *(2004 to 2019)*	• LSTM was used to forecast WQ parameters and then fed into EFDC for spatiotemporal simulation of chlorophyll-α. The resulted LSTM-EFDC showed Nash-Sutcliffe efficiency (NSE) > 0.8 for 5 days ahead prediction.
(AlDahoul et al., 2021)	SS	Johor River Basin, Malaysia	LSTM	Sediment and river flow discharge at daily interval 3995 sets *(1988 to 1998)*	• The LSTM outperformed ElasticNet LR, XGBoost and MLP in forecasting SS up to 1, 7, 10 and 30 days ahead by maintaining accuracy > 90 % for all cases.
(Zhang and Jin, 2020)	TN and TP	Langfang River, China	AE-LSTM LSTM	Temperature, turbidity, pH, EC, TN, TP, DO, COD and AN	• The AE reduced the input dimension for the LSTM.

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Table 3 (*continued*)

Authors	Parameter	Study Area	Model	Input Data	Research Remarks
				450 sets *(Jul to Dec 2018)*	• The AE-LSTM-spatiotemporal model (with WQ data from nearby stations) demonstrated higher R^2 than AE-LSTM (from a single station only) and LSTM.
(Baek et al., 2020)	TN, TP and TOC	Nakdong River Basin, South Korea	CNN-LSTM LSTM	Water level, TN, TP, TOC and weather data *(Jan 2016 to Nov 2017)*	• The CNN simulated water level based on weather data, where its output was fed into LSTM. The hybrid model well represented the trend of the concentration of pollutants regardless of the season.
(Yan et al., 2020)	TN	Juhe River, China	PSO-DBN-LSSVR DBN	Temperature, pH, DO, EC, turbidity, COD, TP and AN at 4 h interval 1086 sets *(2018 to 2019)*	• The DBN was optimized by the PSO to extract useful features and fed into LSSVR for TN prediction. The proposed PSO-DBN-LSSVR showed lower prediction errors than the LSSVR, DBN and BPNN.
(Yan et al., 2021)	TN, TP and TOC	Luan River, China	CNN-Bi-GRU Bi-GRU	COD, TN and TP at 4 h interval 1360 sets *(Jul 2018 to Mar 2019)*	• The Bi-GRU showed higher accuracy than the standard GRU and LSTM. The proposed CNN-Bi-GRU had higher stability across different stations and outperformed the PSO-DBN-LSSVR proposed by Yan et al. (2020).
(Dong and Zhang, 2021)	Polycyclic aromatic hydrocarbon (PAH)	8 major rivers in Saxony, Germany.	CEEMDAN-VMD-LSTM CEEMDAN-LSTM LSTM	PAH data at monthly interval	• The mutation points in the dataset contributed to uneven data distribution and resulted in inadequate model learning. The CEEMDAN-VMD-LSTM with two-staged decomposition addressed the issue and outperformed the best.
(Lu et al., 2021)	DO	Jinjiang River Basin, China	TS-attention-LSTM LSTM RBF-SVR SARIMA	Temperature, turbidity, DO, COD, NH_3, TP, pH, EC, and weather data at 4 h interval *(Mar 2017 to Aug 2019)*	• The optimum historical time step was reported to be 36. • The proposed spatiotemporal-attention-LSTM (TS-attention-LSTM) accurately predicted DO at 12-time steps ahead with MAPE = 3.36 %.
(Song et al., 2021)	DO	Haihe River Basin, China	Optimized SWT-LSTM SWT-LSTM LSTM	DO at weekly interval *(Jan 2010 to Dec 2016)*	• Synchrosqueezed wavelet transform (SWT) showed better denoising performance than the EMD and VMD. The optimized SWT-LSTM performed consistently at all stations.
(Shu et al., 2021)	TN, TP, NH_3, MnO_4, DO, pH and temperature	Nanfei River, China	GRU-AE GRU	TN, TP, NH_3, MnO_4, DO, pH and temperature at 4 h interval 4560 sets *(Aug 2018 to Sept 2020)*	• Input-decay module was coupled with GRU-AE. The GRU-AE was better than linear and mean interpolation for missing value imputation and outperformed GRU and RF in short-term WQ prediction.
(Lee et al., 2022)	DO	Han River, South Korea	GRU LSTM	EC, TN, TP, SS, TOC, turbidity, temperature and weather data *(2017 to 2019)*	• Both GRU and LSTM quickly reflected the short-term changes in DO thus outperforming the ANN. The GRU showed volatility < 10 % when it was challenged with test data from different periods.
(Jia et al., 2020)	Temperature and streamflow	Delaware River Basin, USA	PG-RGN RGN	Temperature and streamflow 9810 sets (temperature) 206,920 sets (streamflow) *(1980 to 2016)*	• Prior knowledge embedded in the physical model enhanced the learning of latent representation in a recurrent graph network (RGN). The resulted PG-RGN had significant improvement over RGN and process-based model at different seasons and river segments.
(Qiu et al., 2021)	Temperature	Yangtze River, China	LSTM	Temperature, discharge rate and air temperature at daily interval *(1983–2013)*	• The LSTM accurately predicted the mean daily variation in the thermal regime and outperformed the air2stream (mechanistic model), RF and BPNN in forecasting daily river water temperature.
(Sadler et al., 2022)	Temperature and streamflow	Rivers in USA	MT-LSTM LSTM	temperature, weather and streamflow data at daily interval *(1980 to 2014)*	• The MT-LSTM predicted streamflow and temperature simultaneously. The MT-LSTM outperformed the single-task modelling approach and was more stable when applied in sparse data conditions across different seasons.

3.8. Concluding remarks

A prediction model that is solely based on an individual algorithm typically contains only partial information about the object (Singha et al., 2021). Combining the DL model with other algorithms may allow for more comprehensive information to be extracted. Here, we summarise the common hybrid models employed for WQ prediction. As the CNN is capable of extracting time-invariant features, it is commonly hybridized with the other models as a feature extractor. Through the extraction of effective representation and local features from a huge dataset, the CNN improves model computation efficiency, minimizes model complexity and eventually had been claimed to be able to outperform many other algorithms like the PCA and HOPLS when coupled with the DL models. As reported earlier, the CNN was often associated with the LSTM for time series forecasting in different applications and had also outperformed most DL models. The fusion of spatial and temporal information by the CNN and the LSTM module allows for relevant variables information to be fully extracted, leading to its adaptability for complex non-linear systems in WQ prediction. Despite having a longer training time, the hybrid models typically converged with fewer training epochs. It was also found that limited comparisons were made for the CNN-GRU in algal bloom forecasting and WQI time series forecasting, hence future studies are recommended on these gaps.

The recurrent-based LSTM and GRU mainly serve as a temporal forecasting model. In some studies where the LSTM had failed to capture long-term trends, a statistical model like the SARIMA was adopted to capture the linear trend of the WQ parameter for better LSTM prediction. The recurrent-based models were also coupled with process-based mechanistic models to integrate scientific knowledge into the black-box

models. The resulted physic-guided hybrids often showed higher robustness and accuracy in WQ forecasting across different seasons or spatial coverage. It was noted that the GBDT combined weak learners with strong learners and was able to assist the GRU with the 3D DO prediction. Through the unsupervised encoder-decoder process, the AE also performed dimensionality reduction and in-depth features for the LSTM. Even though many types of AE have been proposed, only the conventional AE and SAE have been coupled with the LSTM for WQ forecasting. Other types of AEs could be hybridized with the LSTM and GRU and then be compared with the popular CNN-LSTM and CNN-GRU models. Given that the LSTM typically trains slower than the GRU, however, the performance of the hybrid models was seldom compared, hence future studies may focus on this direction. Combining different denoising algorithms such as the CEEMDAN and the VMD has been reported with better signal decomposition and denoising performance can be achieved. Given that different datasets may require different denoising algorithms, further research and development of creating a guideline on this topic would be beneficiary.

The RBM was seldom used for WQ prediction studies owing to its weak ability in handling sequential data. The stacked RBMs, more generally known as the DBN, are commonly employed for extracting inherent correlations between WQ variables. By having a self-learning capability, the DBN could form hybrids with other DL algorithms to serve as a feature extractor and become a prediction model. As there are more parameters to be trained, researchers may pay more attention to optimizing the DBN structure and consider coupling the DBN with different algorithms to improve its DBN performance such as event-triggered algorithm, recursive-time algorithm, statistical model and so forth.

The attention algorithm was coupled with DL models to allocate more computation power to the important parts of the data. Without introducing abundant parameters, the attention algorithm helps to select key input features for target prediction and improve training efficiency. Several attention mechanisms have been identified and comparisons should be made for the different WQ prediction studies using datasets collected from different water environments. In the event of insufficient training data, a TL algorithm was hybridized with DL models to realize model training. Knowledge transfer has been adopted in optimizing the DBN network parameters and transferring knowledge learned in a large dataset to a small dataset. Most studies hybridized with the TL algorithm have reported a positive result where the fine-tuning would shorten training time and require fewer data than training everything from scratch. Nevertheless, the model with the TL typically demonstrated higher generalization ability with higher prediction accuracy and is highly recommended if the target dataset is too small for sufficient DL training.

4. Application of deep learning algorithms for WQ prediction

4.1. Rivers

Many mechanistic models have been developed to forecast WQ in rivers including the River and Stream Water Quality Model (QUAL2K), Water Quality Analysis Simulation Program (WASP) and so forth (Ahmed, 2017; Zhu and Heddam, 2020). However, expert knowledge is needed to pre-set the boundary conditions and system characteristics (Cox, 2003). These mechanistic models also take upstream boundary flow conditions, water level, pollutant concentration and other parameters in time series format for dynamic modelling and this further increases the amount of data required. Some complex hydraulic models even seek boundary conditions at the bottom of the river system and the data of each reach, whenever there are some influencing tributaries or effluent discharge points.

Apart from the mechanistic model, traditional statistical models were used for river WQ parameter time series forecasting. Given that the WQ parameters are influenced by complicated factors, conventional models based on linear distribution cannot comprehensively consider the influence of these factors (Bi et al., 2020). Non-linear ML models usually performed better than the statistical models, but their shallow architecture was unable to extract inherent features for better model training, especially when the datasets were sufficiently large and complex (Lu et al., 2021). Owing to this, the DL-based models with recurrent architecture, such as the LSTM and the GRU become the most popular time series modelling tools in river WQ forecasting (Barzegar et al., 2020; Xu et al., 2021). Table 3 summarizes the studies conducted in river WQ forecasting using DL algorithms. Researchers reported that DL algorithms typically outperformed the traditional statistical models and shallow ML models (Dong et al., 2019; Bi et al., 2020; Lu et al., 2021). From the summary of past studies listed in Table 3, it was found that the DO is the most frequently studied parameter, followed by the organic pollutants parameters like COD, BOD, chlorophyll-α and so forth. Some researchers adopted the COD and BOD as a comprehensive index to reflect the WQ evaluation of rivers (Ye et al., 2019). Research concerning nutrients, heavy metals or other pollutants is still limited. Most studies focused on short-term prediction using hourly data, probably since the WQ variable can be easily influenced by different parameters. Further studies can be conducted to improve the long-term river WQ prediction by developing a robust model with a stronger ability to capture short-term and long-term temporal patterns. The DL models could be hybridized with denoising algorithms, which aim to smoothen the input signal for better model learning and prediction. The TL algorithm could also be hybridized with DL models for river WQ forecasting; for instance, pre-training the model using data from the upstream station, then fine-tuning and applying it to the downstream station for WQ forecasting.

The water quality index (WQI) is a single number designed to turn the complex WQ data structure into a simpler description to ease water management. The process of obtaining the WQ parameter data through sampling at different points would be very costly and time-consuming (Chia et al., 2021). Nevertheless, calculating the WQI via an empirical equation is a tedious process and the computation of the WQI would not be possible in case of any missing WQ parameter. Owing to these reasons, researchers have been adopting the ML models such as the LSSVM (Chia et al., 2021), ANN (Gazzaz et al., 2012; Hameed et al., 2017) and hybridized-ELM (Abba et al., 2020) for river WQI prediction; whilst the applications of DL algorithms for WQI prediction have rarely been reported. Therefore future studies that focus on such directions would be highly recommended and welcomed.

4.2. Lakes

For lake pollutants forecasting, Lu et al. (2022) reported that the LSTM performed better when the historical data for both inflows and internal loads were provided. Based on our literature search, the majority of the studies used DL models to predict harmful algal bloom and relevant WQ parameters such as chlorophyll-α and nutrients. Table 4 presents the past research on lake WQ prediction using DL algorithms. It was generally believed that a chlorophyll-α concentration higher than 10 μg/L or a density of cyanobacteria above 2×10^7 cells/L was the critical value for cyanobacterial blooms (Oliver and Ganf, 2006). The water temperature of 26 °C seemed to be the most suitable temperature for the aggregation of cyanobacteria and the formation of bloom (Deyu et al., 2008). Harmful algal bloom prediction is a difficult task as most of the aquatic system data are sparse and incomplete, thus hindering the development of a good multivariate predictive model.

Traditional chlorophyll-α modelling was performed based on the hydrodynamic and WQ models, such as the EFDC and CE-QUAL-W2 which require expert calibration (Lee et al., 2020b). With the rapid growth of AI, various ML techniques have been reported for their application in the prediction of chlorophyll-α, including ANN (Li et al., 2017b; Lee and Lee, 2018), ANFIS (Loc et al., 2020), SVR, DT (Barzegar et al., 2020) and so forth. One major assumption in the ML approach is

Table 4
Past Research on Lake WQ Prediction using DL Algorithms.

Authors	Parameter	Study Area	Model	Input Data	Research Remarks
(Barzegar et al., 2020)	DO	Small Prespa Lake, Greece	CNN-LSTM CNN LSTM	Temperature, pH, ORP, chlorophyll-α, EC, DO at 15 min interval *(Jun 2012 to May 2013)*	• The LSTM was better in predicting higher levels of pollutants while the CNN was better at lower-level ones. The proposed CNN-LSTM captured both levels accurately and outperformed the stand-alone counterparts.
(Wang et al., 2017)	DO	Taihu Lake, China	LSTM	DO and TP 657 sets *(2000 to 2006)*	• The LSTM showed higher accuracy and generalization than BPNN and online sequential ELM, typically when the data has significant fluctuations.
(Zhu et al., 2021)	DO	Taihu Lake and Yixing Lake, China	TL-Bi-LSTM Bi-LSTM	Temperature, algae, pH, turbidity, chlorophyll-α, EC and DO time series at 10 min interval 25,000 sets	• The Bi-LSTM was pre-trained using large datasets in Taihu Lake and applied in Yixing Lake with limited data. The resulted TL-Bi-LSTM out-performed the TL-MLP and SVM with a shorter training duration.
(Tian et al., 2019)	Algae	An estuary reservoir in eastern China	TL-LSTM TL-RNN TL-ANN	Temperature, pH, ORP, EC, turbidity, DO and chlorophyll-α at 5 min interval 1440 sets *(Jun to Sep 2015)*	• The TL has outperformed parameter norm penalties and the dropout method in improving model generalization ability. • TL-ANN stood as the best model in chlorophyll-α dynamic forecasting.
(Choi et al., 2019)	Algae	Daechungho Lake, South Korea	CNN-log-SMOTE CNN-SMOTE CNN-log CNN	Temperature, pH, EC, DO, TOC, TN, TP and chlorophyll-α 1295 sets *(2012 to 2018)*	• The log-transformation altered the skewed distribution of synthetic data and improved the CNN-log-SMOTE model in predicting chlorophyll-α at 7 days ahead.
(Liang et al., 2020)	Algae	Data obtained from EFDC simulation	LSTM	Temperature, chlorophyll-α, TP, TN, AN and phosphate at daily interval	• The RF identified the key drivers of the performance of LSTM to be temperature, chlorophyll-α and TP. The LSTM with five hidden layers could simulate the forecast capacity of EFDC.
(Lee et al., 2020b)	Algae	Daecheongho Lake, South Korea	CNN CNN + HOPLS CNN-LSTM CNN-LSTM-HOPLS	Temperature, pH, DO, EC, TOC, TN, TP, chlorophyll-α and auxiliary at daily interval *(2012 to 2018)*	• The CNN-LSTM outperformed CNN in forecasting chlorophyll-α 5 days ahead. • The CNN-LSTM-HOPLS trained with auxiliary data was the best performing model as the HOPLS effectively captured relationships among input features.
(Wang et al., 2020b)	Algae	Taihu Lake Basin, China	Recursive-time DBN DBN LSTM	TN, DO, temperature and algal density 5016 sets *(2009 to 2010)*	• Recursive-time DBN outperformed DBN and LSTM in predicting algal density.
(Yu et al., 2020b)	Algae	Dianchi Lake, China	WDTD-LSTM-WMF LSTM GA-LSTM SVR-PCA-BPNN	Temperature, pH, DO, permanganate index, COD, BOD_5, AN, TP, TN, Hg, Pb, volatile phenol, oil, Secchi depth and chlorophyll-α 960 sets *(2005 to 2012)*	• The chlorophyll-α data were denoised by wavelet domain threshold denoising (WDTD) while the wavelet mean fusion (WFM) performed data optimization. • The proposed WDTD-LSTM-WMF model out-performed LSTM, GA-LSTM (Jiabin et al., 2018) and SVR-PCA-BPNN (Yang et al., 2018).
(Song and Zhang, 2020)	Turbidity	QingCaoSha Reservoir, China	LSTM	Turbidity, water level, wind direction and wind speed at 3 h interval 2448 sets	• The LSTM converged its prediction error below 0.05 for 8-steps ahead forecasting. • The LSTM model performs better when actual wind field data are provided.
(Zhang et al., 2021)	DO and NH_3	WuLiangSuHai Lake, China	Attention-CNN-LSTM, CNN-LSTM	Topographic, water flow, weather data, NH_3 and DO at 3 h interval *(Jan 2014)*	• CNN-LSTM showed a loss of information when dealing with spatial features. The attention-CNN-LSTM was faster than the hydrological model (MIKE21) in simulating the future trend of WQ.
(Yu et al., 2022)	DO and NH_3	Poyang Lake, China	EWT-FCM-Bi-GRU Bi-GRU GRU CNN-LSTM	DO and ammonia at daily interval *(Oct 2017 to Apr 2020)*	• A hybrid model (EWT-FCM-Bi-GRU) based on empirical wavelet transform, fuzzy C-means clustering and Bi-GRU demonstrated the lowest MAPE at 4.6 % and outperformed other DL models.
(Li et al., 2021b)	Cyanobacterial algal bloom	Taihu Lake, China	ARIMA-LSTM	Chlorophyll-α, pH, TN, TP, algae density and DO at 30 min interval *(Jan 2017 to Dec 2018)*	• The hybrid ARIMA-LSTM model was better than standalone LSTM and ARIMA in capturing high extremes and local fluctuations.
(Lu et al., 2022)	TN and TP	Taihu Lake, China	LSTM	Chlorophyll-α, pH, TN, TP, DO, COD, temperature, Secchi depth, weather data, flow data and MODIS data *(2011 to 2018)*	• The LSTM outperformed RF and GBRT in simulating TN and TP in the lake. • The LSTM that received inflow and load data outperformed the LSTM that received inflow and internal loads individually.
(Cao et al., 2022)	Algal bloom area forecasting	Taihu Lake, China	CNN-LSTM CNN LSTM	Temperature and weather data at daily interval MODIS data *(2000 to 2019)*	• Algal bloom and weather data were fed to CNN, where the extracted features were used by the LSTM to predict algal bloom. The CNN-LSTM model outperformed the standalone CNN and LSTM models.
		Korattur Lake, India	CNN		

(*continued on next page*)

Table 4 (*continued*)

Authors	Parameter	Study Area	Model	Input Data	Research Remarks
(Prasad et al., 2022)	WQI prediction and classification		LSTM AutoKeras model	Turbidity, TDS, COD, pH, phosphate, iron, nitrate, sodium and Cl^- 5000 sets *(2009 to 2019)*	• SMOTE has significantly improved the classification performance of all models • The CNN and AutoKeras models both showed similar performance, but the latter eliminated the effort of searching for the appropriate DL model.
(Read et al., 2019)	Lake water temperature at multiple depths	Mendota Lake and Sparkling Lake, USA	PG-LSTM LSTM	Temperature and weather data at daily interval (*1980–2018*)	• Energy conservation was introduced as a process-based loss term in process-guided (PG)-LSTM. The resulted PG-LSTM had lower RMSE and higher stability than the LSTM and General Lake Model.
(Willard et al., 2021)	Temperature	Lakes in Upper Midwestern, USA	TL-PG-LSTM	Temperature and weather data 20,880 sets collected from 450 lakes (*1980–2019*)	• The TL strategy was applied to train the PG-LSTM model in some well-monitored lakes and applied to unmonitored lakes. • The resulted TL-PG-LSTM model had significant improvement over the PG-LSTM and process-based model. • The TL-PG-LSTM model that trained on 9 source lakes (ensemble modelling) showed lower RMSE.

that the training and testing data must be of the same distribution, which may not hold for chlorophyll-α prediction. The process of algal bloom in rivers and lakes is highly non-stationary and non-linear (Wang et al., 2020c). The prediction accuracy of certain ML models gradually decreases over time when the model is used to predict chlorophyll-α outside the training sample dataset (Tian et al., 2019). As shown in Table 4, the DBN and its hybrids remain among the most popular DL algorithms for chlorophyll-α forecasting and harmful algal bloom prediction.

Given that algal bloom is a rare event, the chlorophyll-α dataset would be highly skewed and synthetic data could be used to solve the data imbalance by enlarging the minority dataset. Choi et al. (2019) showed that the synthetic minority oversampling technique (SMOTE) improved the performance of the CNN in predicting algal bloom. The variation of chlorophyll-α concentration is seasonal and is affected by many parameters such as water temperature (Yu et al., 2020b), wind speed (Ma et al., 2015), nutrients and so forth. Knowing that the WQ parameters could be influenced by the weather data, it is recommended that meteorological sensors be installed in the vicinity of the WQ monitoring stations so that auxiliary weather data could be used to improve model performance. It has been reported that wind field data played an important role in forecasting the turbidity of lake water (Song and Zhang, 2020). TL and remote sensing technology could be adopted for lakes where accessibility poses a problem. The model could be pre-trained using *in situ* data and calibrated satellite images, and then fine-tuned and applied for WQ forecasting at the targeted lake.

4.3. Coastal zones

The release of agricultural and sewage runoff into the coastal regions has resulted in severe coastal algal blooms and has been regularly reported in China (Su et al., 2021; Zhang et al., 2016), South Korea (Kown et al., 2018; Lee et al., 2020a), Hong Kong (Deng et al., 2021), United States (Karki et al., 2018; Yu and Shen, 2021) and many other countries. These algal blooms have devastating effects on coastal fisheries and mariculture practices on the open seas. As different coastal areas have their dominant factors with highly complicated non-linear characteristics, scientists are unable to attribute algal blooms to any specific factor. Owing to the mechanism causing algal blooms to be unclear, predicting algal blooms based on theoretical dynamic models has been rather difficult, often demonstrating poor generalization performance. With the establishment of a marine water monitoring system in many countries, data-driven approaches that handle non-linearity and complex relationships between parameters have become a better option for predicting algal blooms. As shown in Table 5, the DBN and LSTM and their hybrids are most commonly used in coastal WQ forecasting.

Typically, researchers have to cover a wider region in coastal WQ forecasting to ensure that their predictions are representative of the whole study area. It is rather difficult to gain a synoptic view of the spatial extent and temporal variation of WQ for such a vast area, thus *in situ* monitoring remains a challenge to cover such a large-scale event like algal blooms or marine water pollution. Consequently, satellite remote sensing is used as a supplementary algal bloom observation tool with some algorithms developed based on ocean color data from optical satellites (Kown et al., 2018; Karki et al., 2018). Most studies focused on the optically active variables, such as sea surface temperature, SSS, chlorophyll-α, and so forth. In contrast, fewer studies were conducted on the optically inactive variables such as COD and BOD, probably due to insufficient *in situ* sampling coverage for spatial prediction in coastal areas. Unlike the optically active parameters which could be determined from remote sensing images, the detection of the optically inactive COD and BOD parameters relies wholly on physical sensors installed at sampling sites. However, these optically inactive parameters are also crucial for determining marine WQ, hence further research on the optically inactive parameters from remote sensing images is highly recommended.

4.4. Aquaculture ponds

The aquaculture is a dynamic system with the WQ intensely affected by external factors owing to the open environment of the pond. Mariculture, in turn, is a subset of aquaculture whereby seawater is used for farming marine organisms. For indoor recirculating aquaculture systems which are mostly shielded from the outer environment, only several parameters were deemed significant, including temperature, pH, conductivity, turbidity, and AN. DO remains the most essential factor for the survival of aquatic creatures where it directly affects the feed intake, bait conversion rate and disease resistance of aquatic products (Cao et al., 2020). Stunted growth may be observed among the fishes and diseases can easily outbreak when the DO is low. A sharp decline in the DO caused by WQ deterioration in Hainan Xincun Port has resulted in large-scale fish kill in 2017 (Hu et al., 2020b). In a smart aquaculture farm, artificial aeration helps to maintain DO levels by setting the aerator to commence automatically whenever the DO level falls below the threshold. Given that oxygen diffusion is a slow process, early prediction of DO changes would thus be advantageous so that the aeration rate could be adjusted simultaneously.

Prediction of DO in an aquaculture environment falls into two categories: (i) single parameter prediction that predicts DO based on historical DO data without considering the correlation with other parameters (ii) multi-parameter prediction that fully considers the complexity of DO changes with other correlated WQ parameters and meteorological factors. Most aquaculture monitoring systems adopt multi-parameter prediction and consider DO, water temperature, water

Table 5
Past Research on Coastal WQ Prediction using DL Algorithms.

Authors	Parameter	Study Area	Model	Input Data	Research Remarks
(Zhang et al., 2016)	Algae	Zhejiang Coast, China	DBN RBM	Salinity, COD, DO, chlorophyll-α, nitrate, phosphate, acid nitrate, AN, pH, temperature and silicate at biweekly interval *(2008 to 2012)*	• The DBN showed higher generalization and captured the fluctuation trend better than BPNN, typically at the peak of algal cell density where algal bloom occurs. • An optimal DBN architecture of 12–11–11–11-1 was built with the best learning rate reported to be 0.001.
(Qin et al., 2017)	Algae	Zhoushan and Wenzhou coastal areas, China	ARIMA-DBN DBN	pH, DO, COD, NH_3, temperature, salinity, silicate, nitrate, nitrite, and chlorophyll-α at weekly interval 6200 sets *(2008 to 2014)*	• DBN showed a higher R-value than ARIMA in forecasting chlorophyll-α and COD. • The ARIMA-DBN outperformed DBN, BPNN and ANN in forecasting red tide biomass time series.
(Marndi and Patra, 2020)	Algae	Arabian Sea	MW-LSTM LSTM	SSS, sea surface height and temperature at 5 days interval *(2004 to 2016)*	• A moving window (MW)-LSTM, which consists of six LSTMs with an overlapping subset of data, has outperformed the LSTM owing to a better feature extraction strategy due to a moving window setup.
(Rostam et al., 2021)	Chlorophyll-α	Tolo Harbour, Hong Kong	LSTM RNN DNN	Chlorophyll-α, SS, pH, salinity, DO, turbidity, temperature, TOC, NH_3, TP, TN, PO_4^{3-}, nitrite, nitrate and silica 1556 sets *(1986 to 2008)*	• Chlorophyll-α has a low correlation with most of the input variables. • Despite the LSTM outperforming RNN, deep neural network (DNN), RF, DT, SVM, ANN and MLR in forecasting chlorophyll-α, the sudden spike in chlorophyll-α was not captured.
(Wenxiang et al., 2022)	Chlorophyll-*a*	Xiamen Bay, China	LSTM	Temperature, DO, pH, chlorophyll at 30 min interval and weather data at daily interval (*2008 to 2018*)	• A dynamic model was developed by using the LSTM to predict the absolute concentration, change rate and relative change rate of chlorophyll. The dynamic model performed better than that using chlorophyll data alone with higher stability.
(Zou et al., 2020)	DO, AN and pH	Beilun Estuary, China	Multi-time scale Bi-LSTM	pH, DO, COD and AN at biweekly interval and weather data at 3 h interval 2258 sets	• A multi-time scale Bi-LSTM (comprised of three sub-models to consider historical WQ data of different intervals) showed better performance on pollutant level predictions as compared to the conventional Bi-LSTM, LSTM, GRU, RNN and ANN.
(Liu et al., 2021)	DO	Hilo, USA	Hybrid BEGOE model	Temperature, salinity, turbidity, chlorophyll and DO at 15 min interval 1440 sets *(Dec 2016)*	• A hybrid BEGOE model was designed based on the broyden-fletcher-goldfarb-shanno model, Elman network, outlier-robust ELM and ELM, and outperformed the EEMD-LSSVM (Huan et al., 2018) and GA-optimized fuzzy network (Ren et al., 2018) in marine DO forecasting.
(Xu et al., 2019)	COD and AN	Beilun Estuary, China	Muti-feature SARIMA-LSTM	pH, DO, COD, AN and weather data 11,911 sets *(2014 to 2018)*	• A multi-feature SARIMA-LSTM (received auxiliary features) has outperformed the standard SARIMA-LSTM, SARIMA and LSTM in forecasting COD and AN.
(Ma et al., 2020)	BOD	New York City Harbor, USA	DMF-DNN DNN	DO, pH, TP, TSS, EC, salinity, chlorophyll-α, nitrate/nitrite, etc 32,323 sets	• The deep matrix factorization (DMF) coupled with the DNN outperformed DNN, LR, SVR, RF, GBDT and ANN in forecasting harbour water BOD levels.
(Song et al., 2020a)	SSS	South China Sea, Taiwan	CNN-dual path GRU RNN TCN	South China Sea SSS datasets collected from REDOS 7605 sets *(1992 to 2012)*	• A CNN-coupled dual-path GRU has outperformed TCN, GRU, RNN, SVR, ANN and ARIMA in forecasting SSS. • Prediction of SSS would be challenging on longer days ahead and during season changes where SSS values would become unstable and not follow a regular pattern.
(Yu et al., 2020a)	Temperature	East China Sea and Yellow Sea	DGC network GRU LSTM RNN	Temperature at daily interval 6040 sets *(2001 to 2017)*	• The GRU had larger prediction errors near the maximum values of the data. • The GRU and CNN were hybridized as a DGC network for sea surface temperature prediction. The proposed DGC maintained a higher prediction accuracy (>98 %) than the standalone counterparts
(Kim et al., 2020a)	Temperature	Jeollanam coastal area, South Korea	LSTM	Temperature, wind and air temperature at daily interval 2650 sets *(2008–2017)*	• By incorporating wind and air temperature as auxiliary data, the LSTM achieved 4.17 % of MAPE for 7 days ahead of prediction. • The model performance dropped drastically at a higher prediction horizon.
(Jiang et al., 2021a)	Temperature and salinity	Global Ocean near 15°N 160°E	Bi-LSTM LSTM RNN	Argo dataset *(2011 to 2020)*	• The Bi-LSTM had the best performance followed by LSTM, RNN and LSSVM in modelling ocean temperature.
(Jahanbakht et al., 2022)	Temperature	Bohai Sea, South China Sea, and Pacific Ocean	Stacked LSTM-MLP	Temperature and air temperature at daily interval NOAA and Argo datasets (*1981 – 2020*)	• Sea surface temperature and air temperature data were trained in two separate LSTM-MLP models where the McCulloch-Pitts voting ensemble was used to generate predictions. • The addition of air temperature improved model prediction accuracy.

pH, atmospheric humidity, atmospheric temperature, atmospheric pressure, wind speed and wind level in their calculations. Due to the complex relationship between DO and other input variables, statistical models hardly guarantee the accuracy of DO prediction and demonstrate poor self-adaptability. On the other hand, the non-linear ML models lack a memory function of historical data, thus performing poorly in long-term prediction. DL algorithm models were proposed to solve the memory problem of historical data input. Recent studies undertaken in

Table 6
Past Research on Aquaculture WQ Prediction using DL Algorithms.

Authors	Parameter	Study Area	Model	Input Data	Research Remarks
(Lin et al., 2018)	WQI	Zhejiang Shrimp Pond, China	DBN	Temperature, AN, nitrate, nitrite, PO_4^{3-}, chlorophyll-α, pH, salinity, and DO at weekly interval	• For a 5-layered DBN, the learning rate and iteration number were optimized at 0.01 and 1000 where any increment has no improvement in prediction accuracy.
(Dabrowski et al., 2018)	DO	Anonymous prawn pond	LSTM	DO, pH and temperature at 15 min interval 8463 sets	• The LSTM and linear dynamical system (LDS) had a lower RMSE than the ANN and LR in forecasting DO by using pH and temperature as inputs.
(Ta and Wei, 2018)	DO	Mingbo Aquaculture, China	CNN	DO, temperature, EC and pH at 10 min interval 6000 sets (Mar to Apr 2017)	• The proposed reverse understanding CNN captured the fluctuation changes in the data and converged much faster than optimized-BPNN along with higher fitting and prediction stability.
(Huan et al., 2020)	DO	Jintan Fishery Pond, China	GBDT-LSTM LSTM	DO, temperature, pH, and weather data at 10 min interval 4464 sets *(2017)*	• The GBDT-LSTM model outperformed the standalone LSTM, ELM, BPNN, and PSO-LSSVM by demonstrating higher fitting and prediction accuracy.
(Cao et al., 2020)	DO	Aquaculture farm in Zhejiang, China	PCA-Kmeans-GRU PCA-GRU GRU	WQ data (temperature, DO, acidity, alkalinity, and EC) and weather data at 10 and 30 min interval, respectively 10,936 sets (WQ) and 6384 sets (weather) *(Sep to Oct 2018)*	• The PCA-GRU demonstrated higher prediction accuracy than the GRU as well as other PCA-coupled models (LSTM, CNN, ELM and KNN) in multistep ahead DO forecasting. • The PCA-Kmeans-GRU model carried a stable error curve with the least fluctuations when the prediction horizon increased.
(Li et al., 2021a)	DO	An aquaculture crab pond, China	GRU LSTM	Temperature, pH, turbidity, NH, and DO 5190 sets *(May to Jul 2018)*	• Both LSTM and GRU shared a similar performance and outperformed the RNN, ANFIS and BPNN • With fewer gates, the GRU was trained faster than the LSTM.
(Zhu et al., 2019)	DO	Yixing Crab Pond, China	LDO-LSTM LSTM	pH, temperature, turbidity and AN 6871 sets *(May 2017 to Jul 2017)*	• Both LSTM and low DO (LDO)-LSTM performed well for general DO trend prediction, however, the LDO-LSTM showed lower errors in a low DO scenario (DO < 6 mg/L).
(Li et al., 2018)	DO	Anonymous shrimp pond	SAE-LSTM LSTM	temperature, DO, AN, pH and weather data at 10 min interval 2880 sets	• The SAE effectively extracted hidden features to improve the shortcomings of LSTM in handling sparse features. The resulted SAE-LSTM has outperformed the LSTM, SAE-BPNN and BPNN models in multistep ahead forecasting.
(Liu et al., 2019c)	DO	Zhejiang Institute of Freshwater Fisheries, China	RNN LSTM	DO, temperature, weather and soil data at 10 min interval 5006 sets *(Jul to Aug 2017)*	• The attention mechanism improved the model performance DO prediction. • The temporal and spatiotemporal joint attention mechanisms outperformed other mechanisms regardless of the forecasting horizon.
(Eze and Ajmal, 2020)	DO	Mingbo Mariculture pond, China	EEMD-LSTM SAE-LSTM SAE-BPNN LSTM	DO, pH, temperature, AN, nitrogen and salinity 12,852 sets *(2019)*	• The EEMD-LSTM has outperformed SAE-LSTM, SAE-BPNN, LSTM and BPNN models proposed by Li et al. (2018) in short-term (12 h) and long-term (30 days) DO forecasting.
(Ren et al., 2020)	DO	Shandong Aquaculture, China	VMD-DBN CNN	DO, pH, temperature, turbidity, AN and water level at 10 min interval 12,700 sets *(2018 to 2019)*	• The VMD significantly minimized the noise, highlighted the key data features and improved the model stability. • The VMD-DBN has outperformed the CNN, DT and DBN in forecasting DO.
(Liu et al., 2020)	DO, pH and temperature	Anonymous smart mariculture farm	Bi-S-SRU RNN LSTM SRU	Temperature, salinity, pH, DO, and weather data at 5 min interval 23,000 sets	• The Bi-directional stacked SRU model (Bi-S-SRU) has demonstrated higher stability and accuracy for the next 3 to 8 days of prediction at the cost of the higher training duration.
(Cao et al., 2021)	DO	Crab pond in Gaoteng Town, China.	Attention-GRU-GBRT Attention-GRU Attention-LSTM GRU LSTM CNN	Automatic WQ data (DO, chlorophyll, temperature, pH, and turbidity) at 10 min interval, manual WQ data (temperature and DO) at 30 min interval and weather data at 1 min interval 67,552 sets (WQ) 172,512 sets (weather)	• WQ and meteorological data were fused to the GRU to forecast local DO while the attention mechanism improved the quality of hidden layer extraction. • The resulted attention-GRU showed lower MAE and RMSE than GRU, attention-LSTM, CNN and ELM. • The proposed attention-GRU-GBRT accurately predicted DO distribution throughout the pond and outperformed DT, extra tree and bagging models.
(Liu et al., 2019a)	pH and temperature	Hainan Mariculture Pond, China	SRU RNN	conductivity, salinity, pH, DO and temperature 710 sets	• The SRU with an additional cell state was redundant than the RNN but showed higher

(*continued on next page*)

Table 6 (*continued*)

Authors	Parameter	Study Area	Model	Input Data	Research Remarks
					fitting and accuracy in pH and temperature forecasting.
(Hu et al., 2019)	pH	Hainan Mariculture pond, China	LSTM RNN	EC, salinity, chlorophyll, turbidity, temperature, pH and DO at 5 min interval 710 sets	• The LSTM model achieved higher accuracy in forecasting short-term (8.3 h) and long-term (83 h) pH changes. • The LSTM took a shorter training time than that of the RNN model.
(Hu et al., 2021)	Temperature, and DO	A salmon aquaculture pond of unspecified location	Bi-LSTM	Temperature and DO at 1 min interval *(Aug 2018 to Dec 2018)*	• Environmental datasets are inherently connected in both spatial and temporal directions. The Bi-LSTM considered both directions thus outperforming the XGBoost and RF models.
(Zhou et al., 2021)	DO	Crab pond in Jiangsu Province, China.	SW-TCN-TSA SW-TCN CNN-GRU CNN-LSTM CNN GRU LSTM	Temperature, DO, chlorophyll, conductivity, pH, turbidity and NH_3 and weather data at 10 min interval *(Apr 2020 to May 2020)*	• The sliding window TCN (SW-TCN) and trend surface analysis (TSA) was hybridized for 3D DO forecasting. The resulted SW-TCN-TSA has accurately reflected the spatial and temporal distribution of DO and outperformed all other DL models.
(Li et al., 2022)	DO	An unknown aquaculture farm	LSTM-TCN CNN-LSTM LSTM TCN	DO, pH, turbidity, water temperature, water level and AN at 10 min interval 27,077 sets	• LSTM-TCN hybrid model was better than LSTM and TCN in capturing characteristics of historical data at a longer time window thus demonstrating lower prediction error.

aquaculture WQ forecasting have been summarised in Table 6.

Some studies have successfully proven that the accuracy of DO prediction in outdoor aquaculture ponds would be higher if weather data were provided as auxiliary data, since solar radiation, wind speed, atmospheric temperature and relative humidity may affect the rate of oxygen diffusion (Li et al., 2018; Cao et al., 2021). However, the turbidity should be included given that turbid water may reduce sunlight penetration and photosynthesis rate. Providing a variety of factors as input data can allow for consideration of more possible relationships between WQ parameters, but having too much information in the model may increase the computation burden and could also reduce learning efficiency. Dimensionality reduction tools such as the PCA would be useful in reducing the redundancy of the feed data. In most studies, a multivariate time series prediction could be attempted instead of a univariate time series prediction. The low dissolved oxygen (LDO) algorithm proposed by Zhu et al. (2019) is a great idea to improve model sensitivity, where it could be applied for other critical WQ parameters so that any deviation could be detected in advance to safeguard the aquaculture livestock. Indeed, a local point prediction may not reflect the actual DO in the pond. The 3D DO distribution modelling conducted by Cao et al. (2021) and Zhou et al. (2021) provided a better understanding of the spatiotemporal changes of the DO within the area of interest and paved the way for future WQ modelling in different water bodies.

4.5. Wastewater treatment plant (WWTP)

The activated sludge process is widely adopted in municipal WWTP to remove both organic carbon and nitrogen substances. However, the WWTP modelling becomes challenging due to its complex non-linear dynamics with significant disturbances and uncertain time delays. A WWTP is a combination of hydraulic and biochemical processes (Wang et al., 2021). The hydraulic model describes the non-linear dynamic of flow rates and recirculation to the reactor whereas the biochemical model represents the dynamics of the variable evolution process. Interaction between the variables goes along the whole WWTP, leading to a strong coupling effect among parameters and giving rise to uncertainties that increase the difficulties for modelling. The presence of outliers further contributes to ubiquitous uncertainties and ineffective iterations. Under strict environmental regulations, soft-sensing models for WWTP effluent quality are vital. Soft-sensing is an indirect measurement of a target parameter (such as BOD) where the actual value cannot be determined directly using hardware in real-time. It is accomplished by feeding a measured parameter into a pre-defined model to predict the target parameter. A fast and accurate soft-sensing model for effluent quality not only gives an explicit description of dynamics but also provides an alert reference for the pre-planning of necessary remedies or operational parameters tuning action in advance of an abnormal operation occurring.

The ML techniques have been widely applied to existing soft-sensing models on effluent quality in WWTP (Fernandez de Canete et al., 2016; Han et al., 2016). However, ML techniques have a poor capability of extracting deep features from the datasets, which is deemed crucial for an efficient learning process. The absence of this information may lead to a high computation effort with unsatisfying prediction outcomes. To solve this issue, efforts have been made by applying DL algorithms for the prediction of WWTP effluent quality (Guang et al., 2019; Li et al., 2021c; Wang et al., 2021), effluent flow rate (Wang et al., 2019b), power consumption (Cheng et al., 2020) and so forth. The event-triggered learning algorithm has been adopted to ignore non-valuable data pairs and improve model learning, thus converging the model much faster with higher accuracy (Wang et al., 2021).

As shown in Table 7, the COD was the most frequently studied parameter, followed by SS, nutrients and WWTP operational parameters such as inflow and power consumption. The SS and BOD were found to strongly correlate with COD (Guang et al., 2019). The COD is the major monitoring parameter for wastewater quality by reflecting on the quantity of organic matter in the wastewater (Wang et al., 2021). When the COD reaches a threshold value, the water sample becomes rancid as the water self-purification process consumes most oxygen to degrade various organic substances. This phenomenon kills aerobic microorganisms, breaks the natural balance of the environment and leads to the extinction of aquatic organisms. If not treated initially, impurities such as numerous biological corpses will be absorbed by the sludge and thus silt up, which will in turn cause irreversible harm to the whole biosphere in the long run (Guang et al., 2019). Owing to this, the control strategy of aeration systems is mainly based on the influent COD measured by a COD analyser that takes a long time to stabilize. The delay of COD measurement is a major challenge for process control, which practically increases the treatment cost due to the amount of aeration needed and chemicals added (Wang et al., 2019b).

Scholars have developed some dynamic COD forecasting models using DL algorithms and their hybrids. Wang et al. (2019b) adopted a hybrid CNN-LSTM for dynamic modelling of WWTP effluent COD and

Table 7
Past Research on WWTP WQ Prediction using DL Algorithms.

Authors	Parameter	Study Area	Model	Input Data	Research Remarks
(Fan and Zhang, 2020)	BOD	Data from the UCI database	CNN-SVR CNN	Flow, pH, BOD, COD, SS, and conductivity	• The SVR was better than CNN in predicting effluent BOD. • In the hybrid CNN-SVR configuration, the CNN speeded up the computation and resulted in stable performance.
(Guang et al., 2019)	COD	Data from a simulation model	Bi-LSTM LSTM	Feed rate, aeration, effluent BOD, nitrate, SS and COD 335 sets	• The effluent SS and BOD had the highest correlation with effluent COD, followed by aeration, effluent nitrate and feed rate. • The model performance was reported as follow: LSTM > Bi-LSTM > SVM.
(Wang et al., 2019b)	COD	Urban WWTP in Guangdong	CNN-LSTM CNN LSTM	Wastewater temperature, pH, AN, sewage inflow, influent COD and effluent COD at 1 min interval 28,800 sets collected over 20 days	• The CNN-LSTM has outperformed CNN and LSTM in forecasting effluent COD. • Effluent COD mass flow was predicted via two approaches. The direct method with a 2.11 % of prediction error has outperformed the indirect method of 5.13 %, thus, the latter cannot fulfill the industrial application requirements (<5%) for the aeration process control.
(Niu et al. 2020)	COD and SS	WWTP in Guangdong, China	GA-DBN DBN	Feed rate, pH, temperature, DO, influent COD and SS 170 sets	• The GA-DBN demonstrated lower errors with higher fitting than DBN and BPNN. • The number of RBM layers in GA-DBN greatly affects model performance.
(Li et al., 2021c)	COD and SS	Paper mill in Dongguan, China	CNN-LSTM-Attention CNN-LSTM CNN LSTM	Influent COD and SS, flow rate, pH, temperature, DO, effluent COD and SS 170 sets	• A hybrid CNN-LSTM-Attention model (based on spatial information fusion by CNN, temporal information fusion by LSTM and attention module) performed the best in COD and SS forecasting.
(Qiao et al. 2018)	Nutrients	Gaobeidian WWTP, China	SO-DBN DBN	ORP, DO, influent TP, effluent TP, NH_4, pH, TSS, nitrate and temperature 1000 sets *(May to Aug 2015)*	• The proposed self-organizing (SO)-DBN has the largest sparsity degree to ensure sparse representation of the model. The SO-DBN showed a lower error with a shorter training duration than DBN in TP prediction.
(Wang et al., 2021)	Nutrients	Gaobeidian WWTP, China	DBN-EL SO-DBN TL-DBN ESN	ORP, DO, influent TP, effluent TP, NH_4, pH, TSS, nitrate and temperature 2600 sets *(Feb 2018 to Mar 2019)*	• The event-triggered (EL) algorithm updates the weights whenever certain events were triggered. It was more efficient than the time-triggered method and minimized ineffective data learning. • The DBN-EL converged faster without noticeable fluctuation than the DBN and outperformed the SO-DBN, TL-DBN, and ESN in estimating effluent TP.
(Cheng et al., 2020)	Influent temperature, flow and BOD, effluent chloride and BOD, power consumption	Municipal WWTP in Saudi Arabia	EWMA-GRU GRU LSTM Bi-LSTM Smoothed-LSTM Adaptive-LSTM	Influent temperature, influent flow, influent BOD, effluent chloride, effluent BOD and power consumption at daily interval 2557 sets *(2010 to 2017)*	• Exponentially weighted moving average (EWMA) filter smoothened input signal. • The EWMA-GRU outperformed GRU, LSTM and Bi-LSTM in forecasting WWTP operational parameters. • Physical variables are more predictable than the BOD values since complicated biochemical processes were involved.
(El-Rawy et al., 2021)	TSS, COD, BOD, NH_3 and sulphide	El-Berka WWTP, Egypt	LSTM DFB DCB	TSS, COD, BOD, NH_3 and sulphide *(2011 to 2018)*	• BOD, COD and TSS are positively correlated with the flow of WWTP. • The LSTM had lower errors in pollutant forecasting than ANN, deep feedforward BPNN with 4 hidden layers (DFB) and deep cascade-forward BPNN (DCB).
(Jiang et al., 2021b)	BOD, COD, ammonium, TN and TP	Futian WWTP, China	GRU LSTM RNN	TN, TP, temperature, COD, BOD, SS, EC, pH, and NH_4^+ collected 5 times per day and other auxiliary data 1050 sets	• GRU and LSTM had lower RMSE and higher R^2 than RNN, MLP and MLR. • The contribution of each input to WQ prediction could be further explored so that only key input indicators are provided for higher model efficiency.
(Li and Vanrolleghem, 2022)	COD, TSS and ammonia	Quebec City Water Resource Recovery Facilities	GA-LSTM	Flowrate, COD, TSS and ammonia at daily intervals and weather data	• Chebyshev bandpass filter was used to remove noise caused by precipitation. • Autocorrelation showed that lags up to 4 days back had a significant correlation. • The GA-LSTM had lower MAPE and Kullback–Leibler divergence than ANN.

reported that direct forecasting of COD mass flow has a lower prediction error than the indirect method where the COD mass flow was calculated based on the predicted sewage inflow and COD concentration. For multi-parameter forecasting, researchers reported that physical parameters such as water temperature, inflow and power consumption were easier to be predicted, unlike the chemical parameters where complicated biochemical reactions may increase the difficulties of forecasting (Cheng et al., 2020). The WWTP operational parameters (pump rate, aeration, chemical dosage and so forth) could also be fed into the model to optimize cost savings. It is believed that soft-sensing of the WWTP effluent BOD should be thoroughly studied, which aims to replace the traditional BOD measurement that is time-consuming and labour extensive. Apart from these, researchers may also consider more auxiliary data when training the DL models, including drinking water supply data, sewage flow data, weather data and social population data. A study conducted by Jiang et al. (2021b) has successfully proven the effectiveness of using such auxiliary data to improve WQ parameter forecasting in WWTP.

5. IoT and cloud computing for water quality monitoring using DL

With the development of efficient DL models to handle large datasets, continuous monitoring of WQ has become an interesting but challenging area of research. Smart environmental monitoring is made possible with the emergence of modern sensors which are a central part of the ecosystem of the IoT. The IoT comprises devices, including servers, sensors and actuators, that are interconnected wirelessly to the internet for monitoring, controlling and data logging (Ullo and Sinha, 2020). The development of mobile sensors which allows for the spatial variation in point measurement increases the need for data communication among these devices when manpower is unable to track the mobile sensors at all times. This provides real-time analysis to be carried out and the spontaneous adjustment to the sampling strategies (Li et al., 2017a). The data generated by the IoT devices are typically stored in the cloud systems for future retrieval. Consequently, the big data environment for future smart environmental monitoring may approach a stage where the setup is more mature and friendly, not only for monitoring but also for considering statistical anticipation as well.

The merging of IoT-based data collection and DL has provided a new outlook for researchers. This is coupled with the recent reduction in the cost of applying IoT devices for WQ parameter monitoring (Wong et al., 2021). One of the earlier works proved that neural networks with multiple layers could cope with data collected with IoT devices and classify the real-time WQ into three levels (Chowdury et al., 2019). A water potability classification had also been pursued using IoT data-driven DL models (Alipio, 2020). In China, combining the IoT and the LSTM network has turned into one of the innovations in the predictive modelling of WQ. It was reported that the predictions of the WQ parameters by the LSTM network matched well with the actual data collected by the IoT devices. For comparison purposes, the SVM and ARIMA models were deployed for similar functions. However, their performances were far from satisfactory when contrasted with the LSTM network (Liu et al., 2019b). Barzegar et al. (2020) conducted a similar study in Albania, Greece and Macedonia. Three DL models, including the conventional LSTM, the CNN and the hybrid CNN-LSTM, were compared with shallow models such as the SVM and DT. The authors reported that the three DL models performed similarly in predicting DO and chlorophyll using multivariate inputs. Conversely, the kernel-based SVM and binary DT tended to underestimate the two parameters of interest. It is important to note that although the DL models had predicted well generally, however, the results of this study revealed that these models are still susceptible to extreme values or outliers. This would make real-time monitoring a crucial aspect of smart environmental monitoring.

These studies justified the ability of the DL models in handling real-time data from IoT devices and subsequently make meaningful predictions. However, the use of IoT means that data will be collected continuously and the size of the database eventually can become too massive for a typical workstation to process. Some technological giants have offered cloud computing facilities to the public, such as the Amazon Web Service, Google Colab and Microsoft Azure. These enable the researchers and the public to use such platforms to train and test their DL models before field deployment, thus easing substantial costly technical computing requirements.

Performing cloud computing for real-time WQ monitoring and prediction modelling could result in lesser data communication among devices. This eventually could lead to higher efficiency. The current IoT-DL model frameworks require data to be sent from the sensors to both the cloud system (for storage) and the user interface (remote and local) to obtain user inputs. The user inputs then have to move in the opposite direction to the *in situ* actuators to execute any actions. Meanwhile, the user might have to communicate with the cloud system to retrieve data or store historical actions. These steps can be simplified if the DL models are trained using cloud computing and the users are only required to set up rules for plausible executions. Human interference may only occur when unforeseen circumstances happen. This was suggested by Sagan et al. (2020) where the authors used this framework for satellite image processing. The use of cloud computing would ease other innovations of DL models for WQ monitoring. For instance, Zhou (2020) proposed a TL scheme whereby the LSTM network trained in the reference domain was transferred to a target domain with a high ratio of defective data. The transfer involved data pattern transfers, model transfers and task transfers, which can be overwhelming communication between devices. The work can be hastened by integrating IoT devices with cloud computing equipped with DL models.

6. Limitations on water quality prediction

6.1. Data limitation

DL models are data-hungry and require a large dataset for training. In many studies, the authors only performed one-step forecasting using a small dataset. It is recommended to consider sensor networks that offer datasets of larger dimensions and higher frequency. For example, some countries, like Malaysia, have most of the river WQ data collected bi-monthly in the past decades, whereby the dataset could not reflect any short-term WQ changes. Due to limited dataset availability, researchers may only be privileged to train the model using a single monsoon dataset collected over several months. Thus, the model would demonstrate poor generalization ability in predicting WQ across the different seasons. The lacking of weather data may affect the training and performance of the models as some meteorological parameters like wind speed, atmospheric temperature and pressure would have a certain impact on some WQ parameters.

6.2. Poor data quality

Unlike the aquaculture pond and WWTP, the data collection for surface waters can be rather difficult and is also subject to additional errors. For IoT sensors, many issues are reported on the devices; such as power supply issues and network communication problems which inevitably deteriorate data accuracies. The sampling frequency is very low for manual data collection since tremendous efforts would be needed. In any case, specific physical parameters must be measured *in situ* and are subject to human errors arising from the inexperienced operation, faulty sensors, incorrect calibration and other factors. For chemical parameters that must be analyzed in the lab, the WQ parameters may change during transportation and storage. Unfortunately, some data may not be collected due to poor weather and this would invariably influence the quality of the dataset. This would result in many missing data situations or the presence of outlier data points in the dataset. This would also render the need for extra efforts for data processing and subsequently increases the difficulty of model training.

Hence, the incomplete and highly sparse aquatic data often hinder the development of a good multivariate predictive model.

6.3. Lack of integrated real-time water monitoring system

The water supply in some areas can be considered as that of a multi-level water system, covering both the surface water and underground water, for which both WQ have an impact on each other. There is a lack of research work in these areas; for instance, a study addresses WQ impacts on different water levels due to water pollution from a specific water source. Existing published works usually adopted specific ML and DL techniques for water resources in a limited location during a specific period. Hence, there is a huge demand for advanced technologies integrating multi-level water monitoring and evaluation systems using sensor networks and IoT infrastructures. DL techniques are always closely associated with big data (Varadharajan et al., 2022). The deployment of various sensors and IoT devices enables the collection of high-velocity data, which is useful for DL applications in WQ management in two ways. Firstly, more data (with different resolutions) can be used to enhance the accuracy and performance of the DL models through a more rigorous training process. Secondly, the collected real-time data can be fed to an existing DL model for predictive modelling using cloud computing as a precautionary approach. Current IoT applications in most developing countries are still insufficient to emphasize their values throughout the whole stream of DL-based WQ management (Jan et al., 2021).

6.4. Different parameters and threshold limit for comparison

It is almost impossible to make a fair comparison between models developed in different studies, owing to the different datasets adopted and the parameters of interest. Given that a model may work well on dataset A, it may in some probability, not perform similarly on another dataset B. Researchers often face difficulties in making a general comparison for the developed models. Most countries have their standard of discharge limit for different WQ parameters, thus paving the way for more cumbersome and additional difficulties for model comparison. WQ datasets collected from different institutions would contain different WQ parameters, whilst there are limited comparisons among DL models using the same dataset.

6.5. Computation power and hyperparameter tuning

Most DL networks are complicated and contain plenty of hyperparameters to be tuned. Despite some of the variants reducing the number of parameters by merging their gates, the training speed does not improve drastically. This is because the RNN-based models would still have to work through all the sequences, which is a time-consuming process and exponentially worsen for cases with long sequential data. As such, the RNN-based models require intensive computational power, not to mention that the model may fall into a vanishing or exploding gradient problem. Similarly, the DBN networks typically perform better with additional RBM layers and neurons, but this drastically increases computational time cost. Researchers would need to optimize the trade-off between prediction accuracy and training time by designing parameters for their networks. In some studies, dimensionality reduction tools and feature extractors were employed to reduce training complexity, so that only effective and useful information was provided to the model. Critically, there is no standard practice for tuning hyperparameters and researchers are relegated to optimizing the model manually or using an optimization tool.

7. Recommendation for future works

7.1. Model optimization

The structure of the DL networks (number of layers, neurons, gates) can be optimized to reduce model complexity and computational effort. Some researchers suggested extending the number of hidden layers in a multi-layer network for better prediction performance. Typically, the number of the RBMs layers in the DBN could be increased for better feature learning output but at the cost of higher training time. A more effective memory block for the LSTM should be designed to overcome the long training cycle in the LSTM. Critically, future research direction may focus on designing effective self-adjusting strategies for hyperparameters (learning rate, dropout, activation functions, etc) and model structure (historical time step, forecasting horizon, numbers of hidden layers and neurons), where current research mainly adopted optimization algorithms like the ADAM, genetic algorithm, particle swarm optimization and so forth. For the WWTP operation, the trained model can be incorporated with optimization scenarios to target daily optimal WWTP operation, pollutant removal and cost minimization. Similarly, mariculture farmers may integrate pump and aeration rates into the model for multi-feature prediction so that operating parameters could be optimized for lower operation costs.

7.2. Development of powerful hybrid models and ensemble modelling

New hybrid models can be developed by combining the DL, ML, statistical tools and other learning algorithms. Different models possess unique individual characteristics that allow certain inherent features to be extracted, hence most of the hybrid models outperformed their standalone counterparts. For instance, the combination of the CNN-LSTM allowed for both spatial and temporal features to be captured for better prediction performance. An unsupervised algorithm like the AE could be used as an automatic feature extractor. The DL networks may be improved with other learning algorithms such as event-triggered learning so that model could avoid meaningless iterations. Attention algorithms could be hybridized to allocate more computation power to important data while the ensemble model could combine complementary advantages of multiple forecasting models to achieve better forecasting performance. As the direct ensemble would increase the complexity of the model and lead to overfitting, undeniably, appropriate weight location must be determined so that models can be combined organically. Different ensemble modelling strategies could be considered, either by using different modelling algorithms or using different training datasets, which aim to minimize the generalization error of different base models. Process-based physical knowledge can be embedded into the DL models for better representation of the complex flow and chemical transport dynamics in different water bodies, which paves the way for developing a spatiotemporal WQ forecasting model of higher generalization ability. The interacting WQ parameters driven by the same underlying physical processes can be modelled simultaneously for multi-objective optimization via a multi-task learning approach. By leveraging information from related WQ parameters, the stability of the model could be improved, especially in sparse data conditions.

7.3. Real-time monitoring and data assimilation

There is a lack of integrated real-time monitoring and supervision handling system for the WQ evaluation. Application of sensors network and the IoT for real-time evaluation of WQ would be essential for immediate response of the operators in mariculture and the WWTP. For the lakes and rivers, the authorities should be notified immediately so that remedies can be planned, such as stopping the intake points of water purification plants whenever pollution is predicted. There is a minimal combination of research work on WQ prediction and related pollution control at various levels like running water, underground water and

surface water. Data assimilation can be attempted, for which the trained model integrates data obtained from real-time monitoring sensors, updates and improves itself through continuous iteration. This could be one of the research directions to be followed, aiming to develop a real-time prediction model with low computation complexity, high prediction accuracy and robustness.

7.4. Transfer learning (TL) and synthetic data

In situations where the WQ data collected were insufficient to train a DL model or contained too many missing data points, which would render model training impossible, a TL method could be applied. By having a pre-trained model using a similar or related dataset, the model could learn with lesser data in the actual task during the fine-tuning process. Nevertheless, the computation time can be shortened without compromising prediction accuracy. For example, the continuous monitoring data of River A can be used for model pre-training, which is then applied to predict WQ at River B with limited data upon fine-tuning. Similar works can be attempted for different applications such as transferring a model trained in an upstream station to a downstream station. Without these, training a model using a small dataset would easily lead to model overfitting. Researchers also recommended the Bayesian regularization and ensemble averaging method in the case of a small dataset. Training the DL models with a larger number of hidden neurons and layers could be one of the appropriate solutions. For datasets with extreme skewness, such as in the case of algal blooms, synthetic data could be adopted to increase minority data and solve the data imbalance problem.

7.5. Multi-dimensional data for water quality prediction

Apart from designing the prediction model, future research may focus on the quantity and quality of the data collected, given that all DL models are data-hungry. Smart IoT sensors should be employed to collect more data at fixed intervals so that a multi-dimensional input dataset with higher frequency would improve the extraction of hidden interactions between influencing WQ factors. For the WWTP-related studies, researchers may consider more relevant parameters like precipitation and climate factors over a longer research period, since the concentration of pollutants fluctuates daily. Similarly, the study of surface WQ must consider seasonality as the WQ may change with the seasons. Some researchers opined that hybrid models should be created to forecast the WQ at peak and non-peak times. As the sequence of the environmental variable could be useful for surface WQ prediction, the weather and remote sensing data can be incorporated as auxiliary data. The pooled features (meteorological and geographical) can be used to identify the appropriate parameters to predict the WQ. Multivariate prediction typically works better than the univariate counterpart as the former realizes comprehensive prediction of the water body as a whole. However, this comes with higher model complexity but with the chance of introducing uncertainty to propagate within the model. Researchers should always be cautious when handling multi-parameter datasets and dimensionality reduction tools should be employed.

7.6. Remote sensing spatial prediction and 3D modelling

Remote sensing images can be used for the wide-area WQ prediction involving optically active parameters (such as chlorophyll-α, turbidity, colored dissolved organic matter and so forth) by studying the image color and textures. A better strategy of extracting optically passive parameters such as the COD, BOD and nutrients could be a future research direction. Given that the DO is strongly influenced by surface water temperature, which can be easily interpreted from satellite images, a combination of remote sensing images and environmental variables sequence may predict the DO patterns in surface waters by building a TL-based model. By installing DO sensors at different underwater positions, a 3D spatial DO prediction model could be developed to understand better the DO patterns in different regions of the water environment so that remedies could be planned. Similar work can be extended to other WQ parameters whenever the 3D spatial modelling is deemed necessary since a local point prediction may not represent the entire water environment. Instead of predicting upper-ocean salinity, the 3D spatial modelling of ocean salinity would combine the temporal and spatial information, where the output could be used to infer other variables related to sea salinity like freshwater flux and direction of the ocean current. Successful works can pave the way for other surface water bodies and aquaculture implementation.

7.7. Advanced denoising techniques and dimensionality reduction

WQ and meteorological time series typically contain temporal noises where effective noise reduction must be conducted before model training. More advanced denoising strategies can be studied: for instance, a decomposition algorithm that can adaptively determine the optimal decomposition level should be investigated to replace existing manually-designed methods based on the user's experience. Hence, future studies may focus on multi-level denoising similar to those reported in the literature (Fijani et al., 2019). Bi et al. (2020) even suggested that the DL methods like the stacked AE, RBM and generative adversarial network could be used for signal pre-processing. Given that WQ parameters may have unknown interactions with each other, most researchers tend to feed all collected data for model training, including auxiliary features like weather data. Providing too much invaluable data would burden the training process without improving its prediction accuracy. Thus, data dimension reduction and feature extraction would be helpful to adaptively select the most crucial input features from the context effectively. Dimensionality reduction tools like the PCA and AE and their hybrids are highly recommended. The effects of increasing network layers of the AE for deeper latent features extraction and dimensionality reduction could be further investigated.

8. Conclusions

This review paper serves to cover the importance of the DL models in numerous instances related to WQ management. It began with the introduction of the ANN and RNN models. Researchers attempted to fix the pitfalls of the RNN model in solving massive time series by introducing the different variants of the LSTM, GRU, CNN, TCN and DBN models. The characteristics of the different DL models were compared and contrasted.

Many studies claimed that the performance of the DL models could be improved with various hybridization techniques. These hybridization techniques come in various forms including of the combinations of different base DL models (e.g. CNN-LSTM model), process-based hybridization (e.g. PG-LSTM), statistical-based hybridization (e.g. SARIMA-LSTM model), ML-based hybridization (e.g. GBDT-LSTM model), attention-based hybridization (e.g. attention-GRU-GBRT model) and TL hybridization (e.g. TL-LSTM model). Evidently, from the perspective of WQ management, the hybrid DL models were found to have outperformed the conventional DL models, in terms of accuracy and fitting based on the results of the available literature. This could be due to the fact that the hybrid DL models excel in various specific tasks, including feature extraction, trend learning and data manipulation.

Throughout the review paper, the application of the DL models in WQ management has been discussed extensively. Most importantly, the scope of this review paper was extended to cover as many types of water bodies as possible, including rivers, lakes, coastal, aquaculture water and even the WWTP. For each water body, the applications of DL models on the specific WQ parameters of interest were discussed separately as the importance of parameters of one particular water body could be irrelevant in another. The WQI, a multi-parameter WQ metric was also explained in conjunction with the application of DL models.

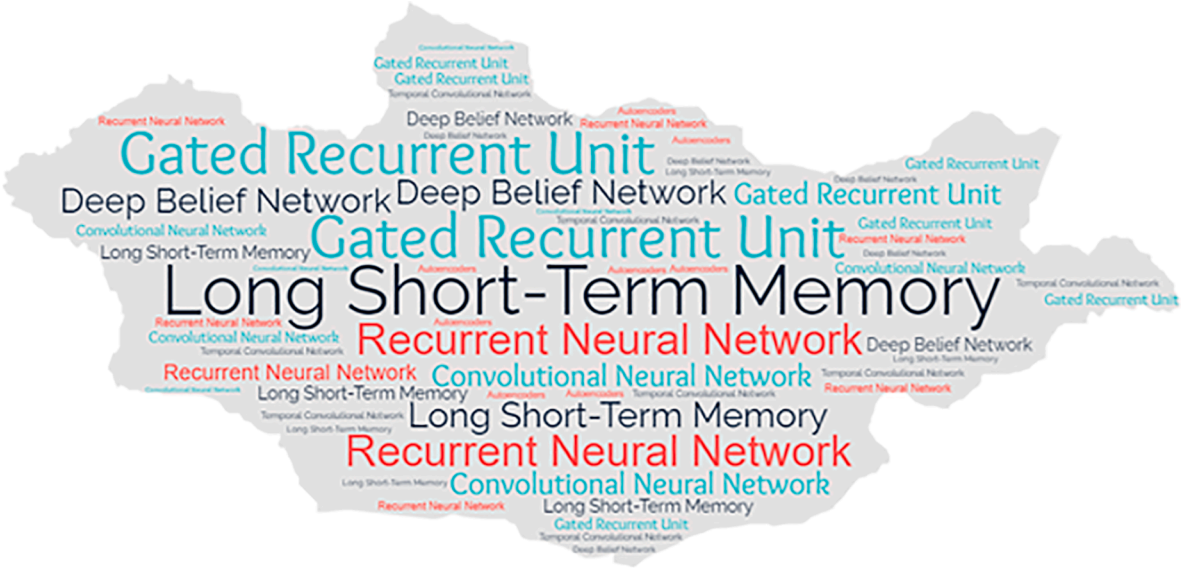


Fig. 6. Word cloud of various DL models adopted for WQ prediction.

Technological advancement has driven the deployment of IoT devices and cloud computing for WQ management alongside the DL models. The review pointed out the ways of merging IoT, cloud computing and DL models, as a real-time monitoring scheme. The inclusion of cloud computing is beneficial when dealing with a gigantic scale of real-time data and higher modelling efficiency. Less human interference is anticipated if the proposed system begins to mature.

Although the DL models possess promising potential in the field of WQ management, several limitations and challenges await to be overcome to reach its full-scale application. At present, the availability of data is still ranked the biggest difficulty, whereby many regions worldwide still lack a proper data collection system and infrastructure. This consequentially leads to the lack of data amount, poor data quality and lack of systems or devices that can monitor the WQ of various water bodies on a real-time basis. Finally, the development of high-performance DL models is always accompanied by a large number of hyperparameters to be tuned. This would pose a severe challenge to the need for computational power.

Lastly, the authors of this review paper suggested several recommendations for future works and studies. For the DL models, upcoming research works can focus on developing lightweight yet powerful DL models by modifying the architecture of the current DL models. Taking it a step further, the researchers can also experiment with different ways of hybridizing the DL models to achieve better performance in WQ management. Data-wise, the authors suggested that the current data collection infrastructures can be further improved with the deployment of more IoT devices and advanced instruments. This can help to mitigate the problem of missing data and at the same time provides more information on the WQ in different dimensions. Alternatively, the data structure can be modified via TL and the generation of synthetic data. The remote sensing images can be utilized to spatially map the WQ to better understand its dynamics. Also, pre-processing (including denoising and dimensionality reduction) shall be useful for extracting valuable and essential information from the available raw data. By improving the techniques, from the aspects of the DL models and data, it is believed that the efficiency and effectiveness of WQ management can be further enhanced, appreciated and boosted to benefit the economy and social welfare of the local communities; the premise which echoes the call of the 6th Sustainable Development Goals of the United Nations. Fig. 6 illustrates a word cloud demonstrating the various DL models adopted for WQ prediction. It is noticeable that the GRU and LSTM are more visually appealing than the other DL models for WQ prediction.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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